ADMINPC01 — Workstation Deployment & Domain Integration

Task 1 – preparation section for Module 4, before the actual administrative tasks begin.

1. Windows 11 workstation deployment

I created the new workstation:

```
Computer: ADMINPC01
Operating System: Windows 11
Initial local account: LabAdmin
Initial membership: WORKGROUP
```

The important concept here is that **ADMINPC01 initially had no relationship with Active Directory**.

`LabAdmin` was a **local account**. It existed only inside ADMINPC01.

---

2. Network configuration validation

ADMINPC01 received its network configuration through DHCP:

```
IPv4 Address:    10.20.20.146
Subnet Mask:     255.255.255.0
Default Gateway: 10.20.20.254
DHCP Server:     10.20.20.254
```

That showed that ADMINPC01 was correctly connected to the `10.20.20.0/24` lab network.

I also tested:

```
ping 10.20.20.10
```

and received replies from DC01.

Learning point: Before attempting a domain join, verify basic IP connectivity between the workstation and the Domain Controller.

---

3. DNS configuration — one of the most important steps

Initially ADMINPC01 was using external Comcast DNS servers.

That allowed Internet name resolution, but:

```
nslookup dc01.benlab.local
```

failed.

Why?

Because Comcast DNS knows nothing about the private Active Directory DNS namespace:

```
benlab.local
```

I corrected the DNS configuration so ADMINPC01 used:

```
DNS Server: 10.20.20.10
```

which is **DC01**.

Then:

```
nslookup dc01.benlab.local
```

successfully returned DC01.

Key learning point

A domain client should use the **Active Directory DNS server**, not a public/ISP DNS server, for AD name resolution.

Internet DNS resolution can then be handled through DNS forwarding/upstream resolution.

This is one of the most important concepts from the entire exercise.

---

4. Active Directory service discovery

I didn't stop after simply pinging DC01.

I tested whether ADMINPC01 could actually discover the Active Directory service through DNS:

```
nslookup -type=SRV _ldap._tcp.dc._msdcs.benlab.local
```

After DNS was corrected, the result identified:

```
port = 389
svr hostname = dc01.benlab.local
```

That is much more meaningful than a successful ping.

Ping asks:

> Can I reach this machine?

The SRV lookup asks something closer to:

> Where is a Domain Controller providing LDAP services for `benlab.local`?

Active Directory relies heavily on these **DNS SRV records** to locate domain services.

---

5. VMware Tools

I installed VMware Tools using the mounted VMware Tools virtual DVD.

I selected:

```
Typical
```

and restarted ADMINPC01 after installation.

This provides the drivers and integration components needed for the Windows guest to operate properly within VMware.

---

6. Joining ADMINPC01 to `benlab.local`

Before joining the domain, System Properties showed:

```
Computer name: ADMINPC01
Workgroup: WORKGROUP
```

I changed membership from:

```
WORKGROUP
```

to:

```
Domain: benlab.local
```

Windows then requested credentials belonging to an account authorized to join computers to the domain.

After successful authentication, Windows displayed the:

```
Welcome to the benlab.local domain
```

message.

The computer was then restarted.

What actually happened?

This wasn't simply Windows changing the word **WORKGROUP** to **benlab.local**.

A computer relationship was established between:

```
ADMINPC01
        ↓
Active Directory
        ↓
benlab.local
```

Active Directory now recognizes ADMINPC01 as a domain computer.

---

7. Local account vs. domain account

Before joining/logging into the domain I had:

```
ADMINPC01\LabAdmin
```

That account exists **only on ADMINPC01**.

After joining the domain, I logged in using:

```
BENLAB\Administrator
```

That account is authenticated by **Active Directory**.

This distinction is extremely important:

```
LOCAL AUTHENTICATION

ADMINPC01
   └── LabAdmin


DOMAIN AUTHENTICATION

ADMINPC01
   ├── DNS → DC01
   └── Active Directory
          └── BENLAB\Administrator
```

---

8. Verification after the domain join

I performed three excellent verification commands.

Identity

```
whoami
```

Result:

```
benlab\administrator
```

This proves I am using a **domain identity**.

Computer identity

```
hostname
```

Result:

```
ADMINPC01
```

This proves which computer I am administering from.

Authentication source

```
echo %logonserver%
```

Result:

```
\\DC01
```

This tells us that **DC01 processed the domain logon**.

Together, those three commands tell a very clear story:

```
WHO AM I?
BENLAB\Administrator

WHERE AM I?
ADMINPC01

WHO AUTHENTICATED ME?
DC01
```

Why ADMINPC01 matters to lab

This is where the project becomes more enterprise-like.

Previously, a lot of administration was performed directly on:

```
DC01
```

But administrators generally shouldn't need to sit directly on a Domain Controller every time they want to manage users, groups, DNS, GPOs, etc.

My architecture is becoming:

```
                ADMINISTRATOR
                      |
                      ▼
                ADMINPC01
          Administrative Workstation
                      |
              RSAT / PowerShell
                      |
                      ▼
                   DC01
```

```
         ┌──────────┼──────────┐
         ▼          ▼          ▼
     AD DS         DNS        GPO
         │
         ▼
             benlab.local
         │            │
         ▼            ▼
     Servers       Workstations
```

That is exactly why the next step—**RSAT**—will matter.

Module 4 – Remote Server Administration

Task 2 – Install and Validate RSAT on ADMINPC01

Objective

Configure **ADMINPC01** as a centralized administrative workstation by installing Remote Server Administration Tools (RSAT) and validating remote management of Active Directory, Group Policy, DNS, and DHCP infrastructure.

The goal is to perform routine administrative tasks from a dedicated management workstation rather than logging directly onto the Domain Controller.

Environment

System	Role	Purpose
ADMINPC01	Windows 11 Administrative Workstation	Centralized management workstation
DC01	Domain Controller / DNS Server	Hosts AD DS and DNS
benlab.local	Active Directory Domain	Enterprise lab domain
10.20.20.10	DC01 IP Address	Domain DNS and AD services

ADMINPC01 was previously joined to the `benlab.local` domain and domain connectivity was successfully validated.

Step 1 – Access Windows Optional Features

On **ADMINPC01**, opened:

Settings → System → Optional features → View features

Searched for:

```
RSAT
```

Windows displayed the available **Remote Server Administration Tools**.

Step 2 – Select Administrative Tools

The following RSAT components were selected:

```
RSAT: Active Directory Domain Services and
Lightweight Directory Services Tools

RSAT: Group Policy Management Tools

RSAT: DNS Server Tools

RSAT: DHCP Server Tools
```

The installation was then started by selecting **Add**.

Step 3 – Verify RSAT Installation

Windows successfully installed all four selected components.

The Optional Features interface reported:

```
RSAT: Active Directory Domain Services and
Lightweight Directory Services Tools    Added

RSAT: DNS Server Tools                   Added

RSAT: Group Policy Management Tools      Added

RSAT: DHCP Server Tools                  Added
```

This confirmed that ADMINPC01 now contained the management consoles required to remotely administer major Windows infrastructure services.

Remote Management Validation

Step 4 – Validate Active Directory Users and Computers

Opened **Active Directory Users and Computers** from ADMINPC01.

Run command:

```
dsa.msc
```

The console successfully discovered:

```
benlab.local
```

Expanded the domain and verified access to the existing organizational structure:

```
BENLAB
├── Groups
├── Servers
├── Service Accounts
├── Users
└── Workstations
```

Existing domain users were also visible from ADMINPC01.

Result

ADMINPC01 successfully accessed Active Directory remotely.

This demonstrated that AD objects can now be managed without interactively logging into DC01.

Step 5 – Validate Group Policy Management

Opened **Group Policy Management** from ADMINPC01.

Run command:

```
gpmc.msc
```

The console successfully discovered:

```
Forest: benlab.local
    └── Domains
        └── benlab.local
```

Existing Group Policy Objects and links were visible, including policies previously created during earlier lab modules.

Result

ADMINPC01 can remotely access the domain's Group Policy infrastructure.

Group Policy administration can therefore be performed from the dedicated management workstation.

Step 6 – Open DNS Manager

Opened the DNS management console using:

```
dnsmgmt.msc
```

Because ADMINPC01 does **not** host the DNS Server role, DNS Manager prompted for the DNS server to manage.

Selected:

The following computer

and entered:

```
DC01
```

The option:

```
Connect to the specified computer now
```

was enabled.

Step 7 – Validate Remote DNS Administration

DNS Manager successfully connected from ADMINPC01 to **DC01**.

Expanded:

```
DNS
└── DC01
    ├── Forward Lookup Zones
    ├── Reverse Lookup Zones
    ├── Trust Points
    └── Conditional Forwarders
```

Under **Forward Lookup Zones**, the existing Active Directory-integrated zones were visible:

```
_msdcs.benlab.local
benlab.local
```

Both zones reported a status of:

```
Running
```

Result

ADMINPC01 successfully established remote administrative access to the DNS Server service running on DC01.

Important Architecture Concept

Installing **RSAT DNS Server Tools** on ADMINPC01 did **not** turn ADMINPC01 into a DNS server.

The architecture remains:

```
ADMINPC01
     │
     │ RSAT / DNS Manager
     ▼
DC01
     │
     └── DNS Server Role
            ├── benlab.local
            ├── _msdcs.benlab.local
            ├── Reverse Lookup Zones
            └── Conditional Forwarders
```

ADMINPC01 contains the **management interface**.

DC01 contains the actual **DNS Server service and DNS database**.

This same general concept applies to the other RSAT components.

Administrative Console Commands

Useful MMC commands validated during this task:

Command	Administrative Tool
`dsa.msc`	Active Directory Users and Computers
`gpmc.msc`	Group Policy Management
`dnsmgmt.msc`	DNS Manager
`dhcpmgmt.msc`	DHCP Manager

These commands allow administrators to launch management consoles directly without navigating through Windows menus.

Key Learning Points

RSAT (Remote Server Administration Tools) allows administrators to manage Windows Server infrastructure from a Windows administrative workstation.

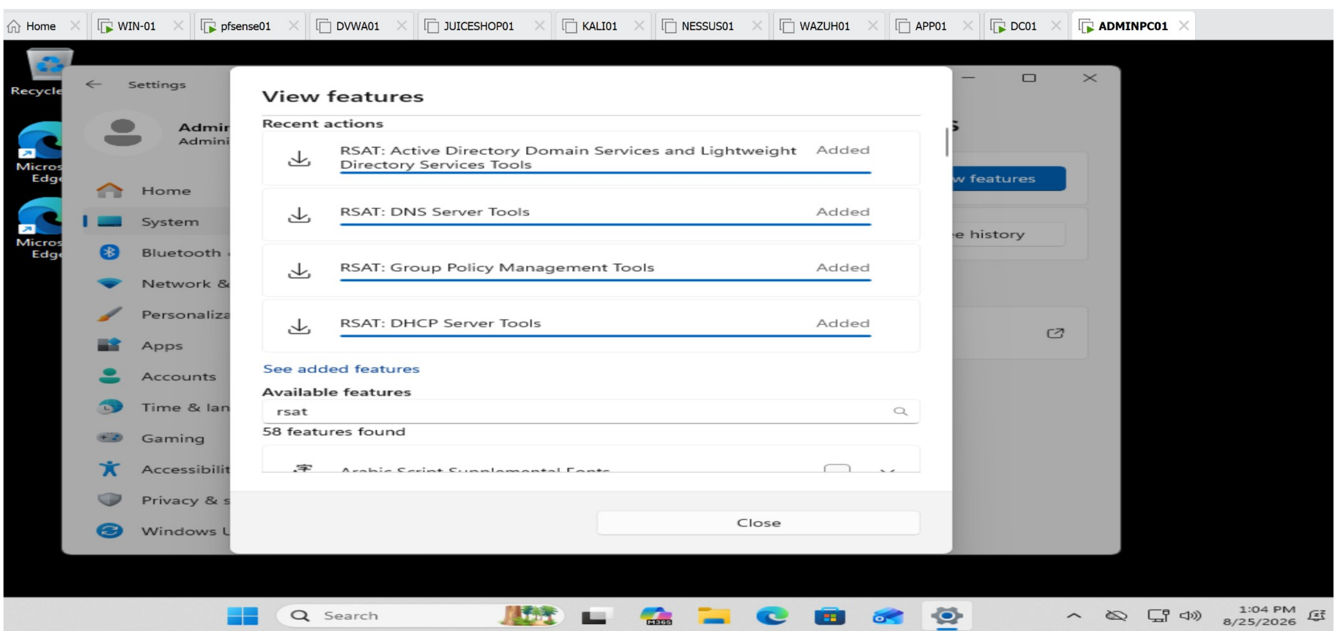
A management workstation does not need to host the server role it manages. For example, ADMINPC01 can contain **DNS Server Tools** while the actual DNS Server role continues running on DC01.

Domain membership and proper DNS configuration allow ADMINPC01 to discover and communicate with Active Directory infrastructure.

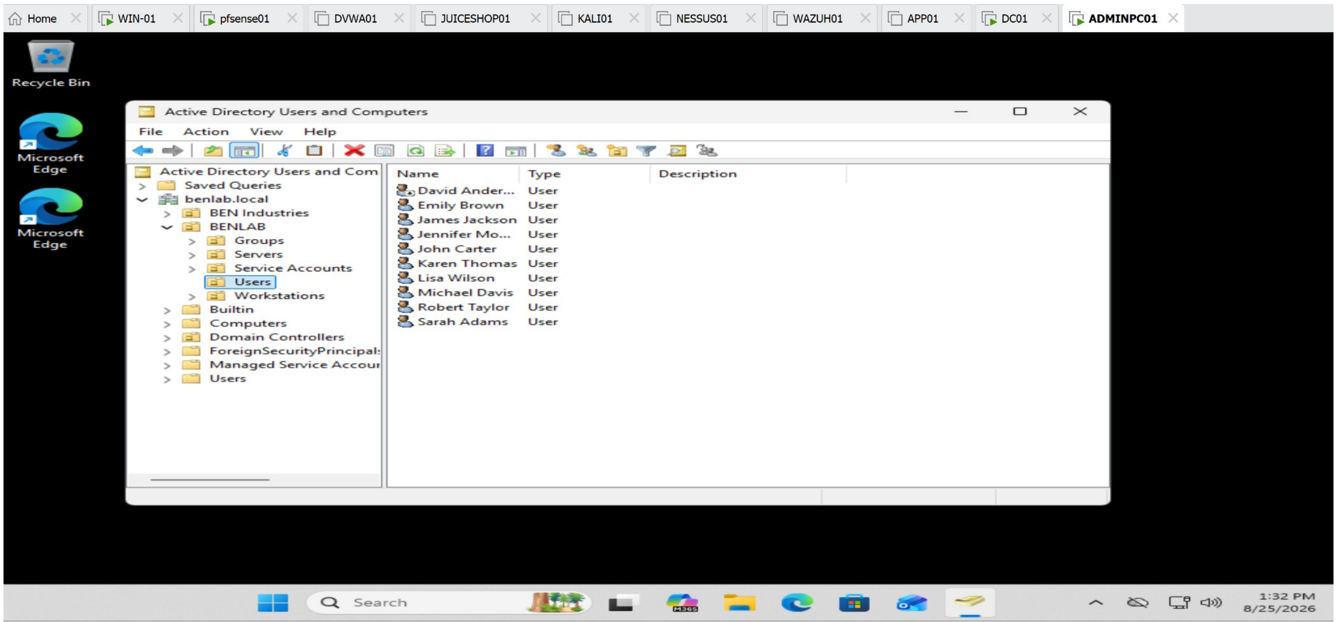
Administrative consoles behave somewhat differently. ADUC and Group Policy Management can automatically discover the Active Directory domain, whereas DNS Manager may require specifying the DNS server that should be managed.

Centralized remote administration reduces the need to log directly into infrastructure servers for routine management.

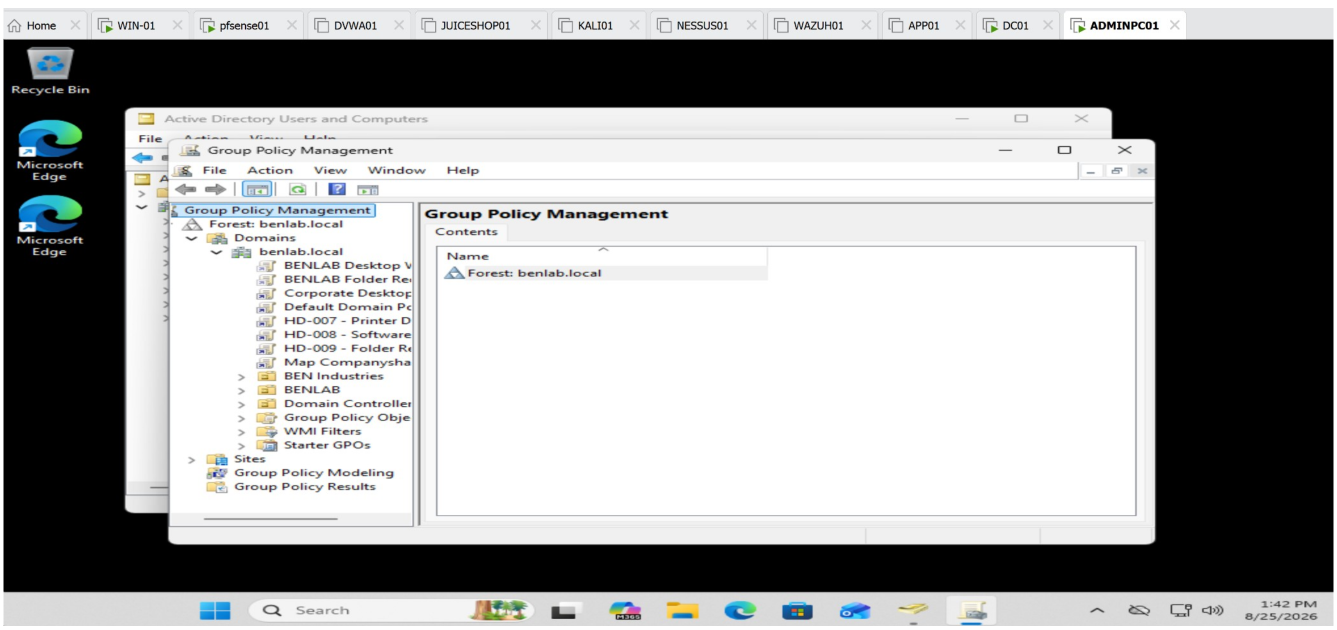
Screenshots / Evidence



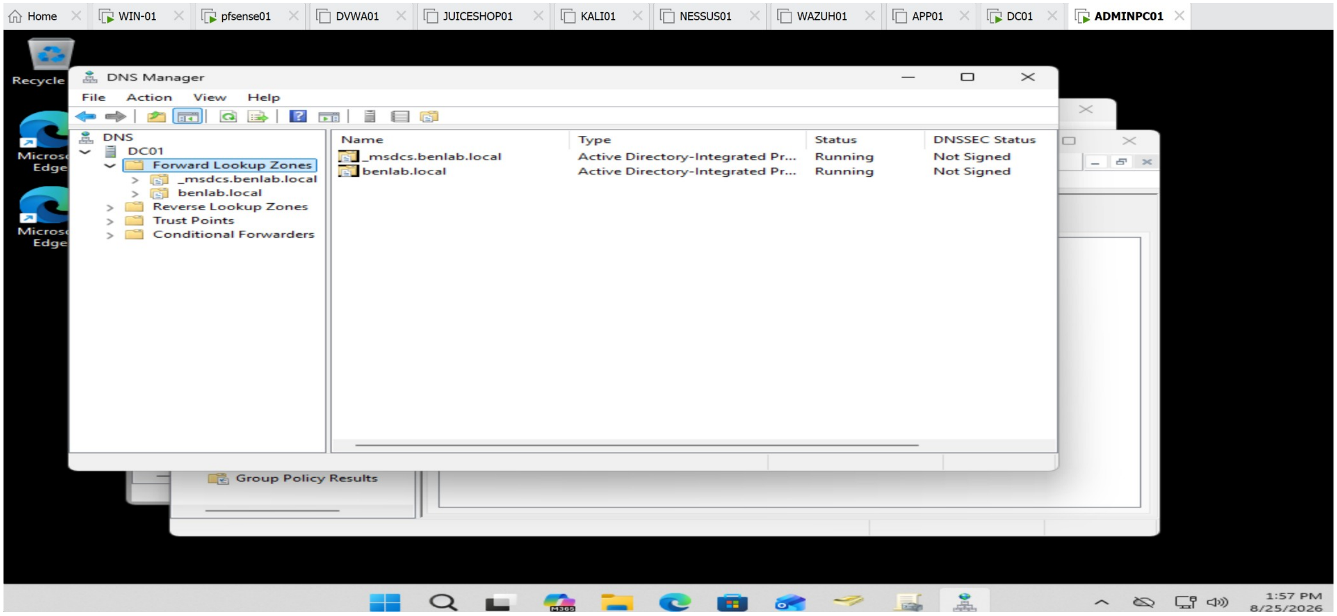
RSAT installation completed — showing the four components as `Added`.



Active Directory Users and Computers from ADMINPC01 — showing the `BENLAB\Users` objects.



Group Policy Management from ADMINPC01 — showing the existing `benlab.local` GPO infrastructure.



DNS Manager connected to DC01 — use the screenshot showing `benlab.local`, `_msdcs.benlab.local`, Reverse Lookup Zones, and Conditional Forwarders.

Task 3 — Configure and Validate a DHCP Reservation

Objective

Configure and validate a DHCP reservation for **ADMINPC01** within the BENLAB environment.

The goal was to provide ADMINPC01 with a predictable IP address while allowing the workstation to remain configured as a **DHCP client**. The reservation was created on **pfSense01**, which provides DHCP services for the `10.20.20.0/24` network.

Environment

- **Domain:** `benlab.local`
- **DHCP Server / Gateway:** pfSense01 — `10.20.20.254`
- **DNS Server:** DC01 — `10.20.20.10`
- **DHCP Pool:** `10.20.20.100 - 10.20.20.200`
- **Client:** ADMINPC01

- **Original DHCP Address:** `10.20.20.146`
- **Reserved Address:** `10.20.20.90`
- **ADMINPC01 MAC Address:** `00-0C-29-CF-0D-EB`

Step 1 — Validate the Existing DHCP Configuration

On ADMINPC01, PowerShell was used to review the workstation's existing network configuration:

```
ipconfig /all
```

The output confirmed that ADMINPC01 was configured as a DHCP client and had received:

- IPv4 address: `10.20.20.146`
- DHCP server: `10.20.20.254`
- Default gateway: `10.20.20.254`
- DNS server: `10.20.20.10`
- MAC address: `00-0C-29-CF-0D-EB`

This established the baseline before modifying DHCP configuration.

Step 2 — Verify the Existing DHCP Lease

The pfSense DHCP lease table was reviewed.

The active lease showed:

- Hostname: `ADMINPC01`
- IP address: `10.20.20.146`
- MAC address: `00:0c:29:cf:0d:eb`

The information matched the configuration reported by `ipconfig /all` on ADMINPC01.

This confirmed the DHCP client/server relationship before creating the reservation.

Step 3 — Create the Static DHCP Mapping

From the pfSense DHCP Server configuration, a **Static DHCP Mapping** was created for ADMINPC01.

The following values were configured:

```
MAC Address: 00:0c:29:cf:0d:eb

IP Address:  10.20.20.90
```

```
Hostname:    ADMINPC01
```

The reserved address `10.20.20.90` was selected because it is within the BENLAB `10.20.20.0/24` subnet but outside the dynamic DHCP pool of `10.20.20.100–200`.

The existing DHCP options were retained:

- DNS server: `10.20.20.10`
- Gateway: `10.20.20.254`
- Domain: `benlab.local`

Static ARP was not enabled because it was not required for the DHCP reservation.

Step 4 — Save and Apply the DHCP Configuration

The static mapping was saved in pfSense.

pfSense indicated that the DHCP configuration had changed and required the changes to be applied.

Apply Changes was selected, and pfSense reported:

> **The changes have been applied successfully.**

This activated the new DHCP reservation.

Step 5 — Release the Existing DHCP Lease

On ADMINPC01, the existing `10.20.20.146` lease was released:

```
ipconfig /release
```

The workstation temporarily lost its DHCP-assigned IPv4 configuration, confirming that the existing lease had been released.

Step 6 — Request a New DHCP Lease

A new DHCP lease was requested:

```
ipconfig /renew
```

pfSense recognized ADMINPC01 by its MAC address and assigned the newly reserved address:

```
IPv4 Address:    10.20.20.90
Subnet Mask:     255.255.255.0
Default Gateway: 10.20.20.254
```

This demonstrated that the static DHCP mapping was functioning correctly.

Step 7 — Validate the DHCP Reservation

A final validation was performed:

```
ipconfig /all
```

The output confirmed:

```
DHCP Enabled:    Yes
IPv4 Address:    10.20.20.90
DHCP Server:     10.20.20.254
DNS Server:      10.20.20.10
Default Gateway: 10.20.20.254
```

The most important validation was that **DHCP Enabled remained set to Yes** while ADMINPC01 received the predictable address `10.20.20.90`.

This confirms that `10.20.20.90` was obtained through the DHCP reservation and was **not manually configured as a static IP address in Windows**.

Validation Result

Task 3 completed successfully. ✅

ADMINPC01 transitioned from the dynamically allocated address:

```
10.20.20.146
```

to the DHCP-reserved address:

```
10.20.20.90
```

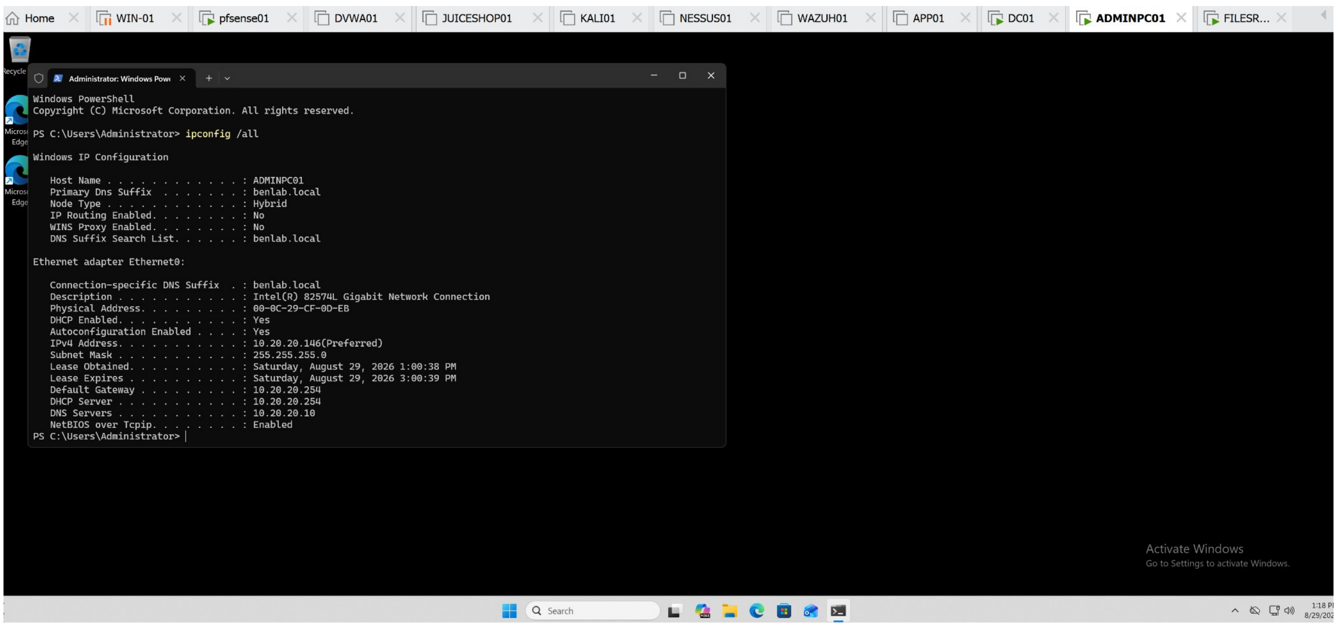
The workstation remained a DHCP client and continued receiving the correct gateway, DNS server, and BENLAB domain configuration.

Key Learning Points

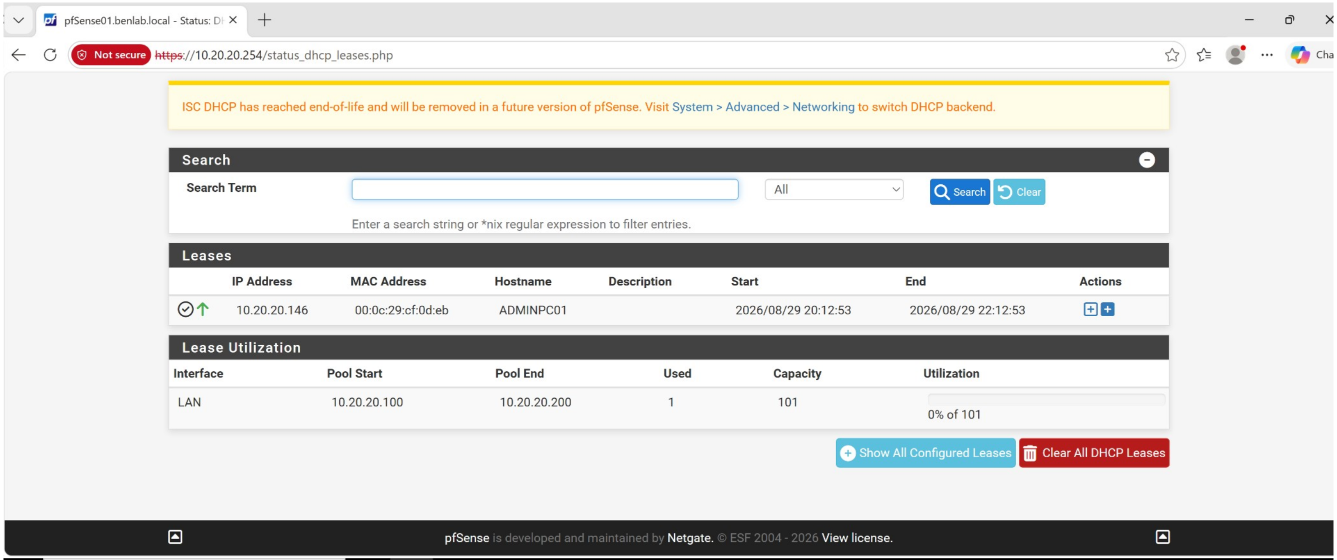
- A **DHCP reservation** associates a specific client identifier, commonly a MAC address, with a predictable IP address.
- A reserved address differs from a manually configured static IP because the client **continues using DHCP**.
- DHCP reservations provide centralized control while giving important devices predictable addressing.
- In this pfSense ISC DHCP configuration, the reserved address was placed **outside the dynamic DHCP pool**.

- `ipconfig /release` relinquishes the client's existing DHCP lease.
- `ipconfig /renew` requests a new lease from the DHCP server.
- `ipconfig /all` provides detailed evidence for validating the assigned address, DHCP server, DNS server, gateway, and whether DHCP remains enabled.
- Comparing the MAC address on the client with the static mapping on the DHCP server helps verify that the reservation targets the correct device.

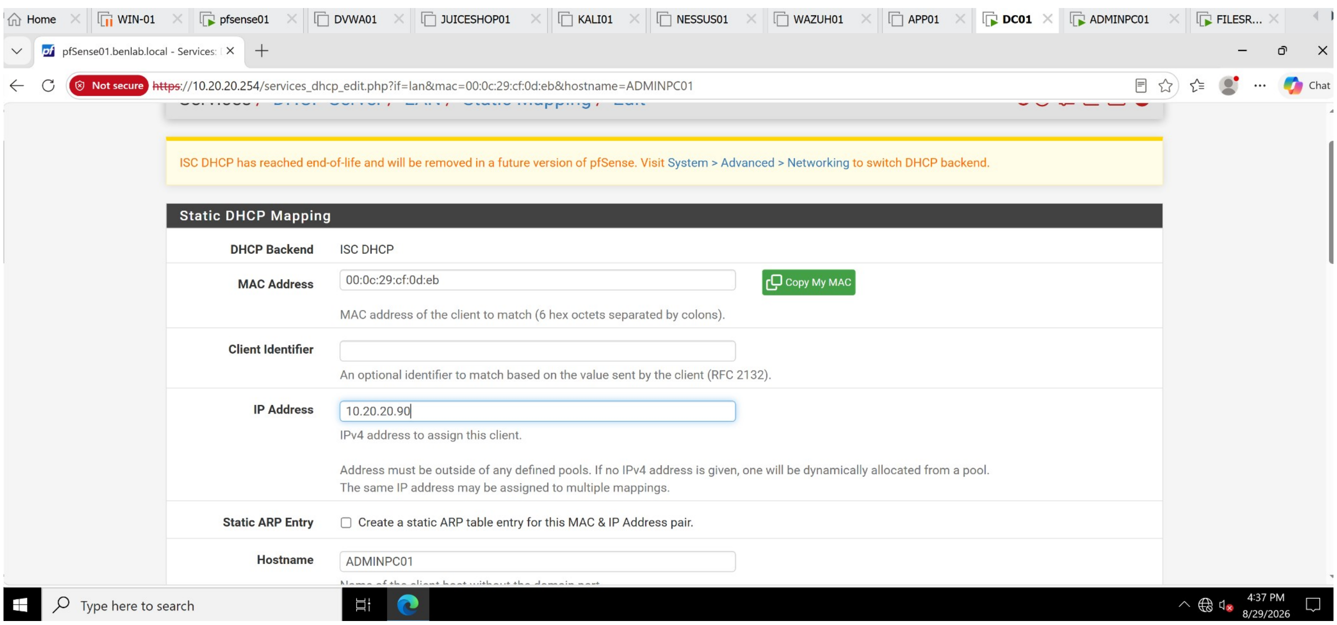
Screenshots / evidence



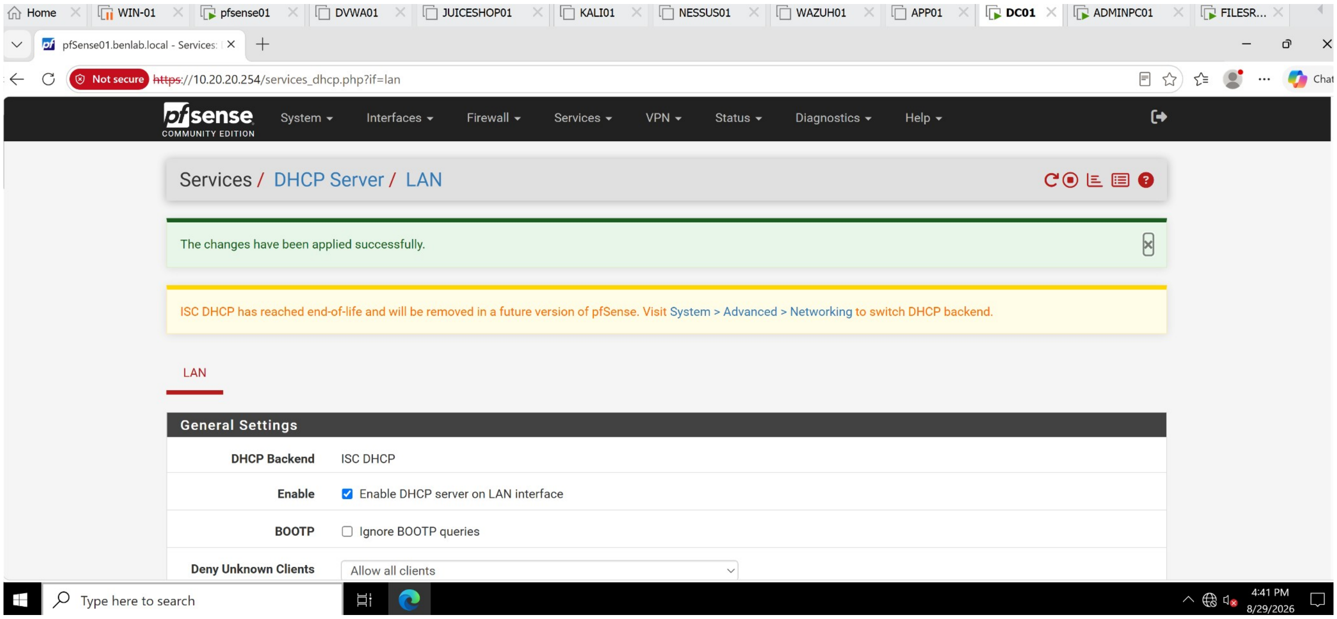
Initial `ipconfig /all` — ADMINPC01 at `10.20.20.146`.



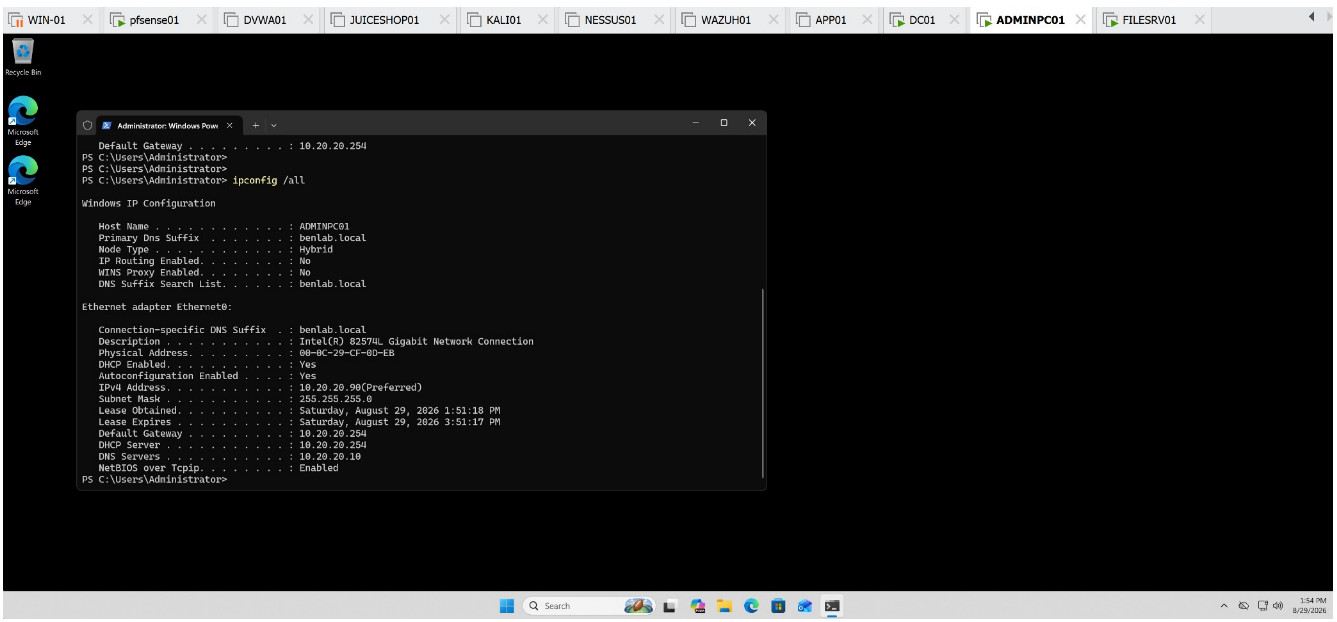
pfSense DHCP lease — showing `.146` associated with ADMINPC01 and its MAC.



pfSense Static DHCP Mapping — MAC mapped to `10.20.20.90`.



pfSense “changes have been applied successfully” confirmation.



Final `ipconfig /all` — DHCP Enabled: Yes + IPv4 `10.20.20.90`.

Task 4 — Configure and Validate DHCP Exclusions

Objective

Configure an excluded address range within the BENLAB DHCP environment to prevent selected IP addresses from being dynamically assigned to DHCP clients.

Because **pfSense01** provides DHCP services rather than Windows Server DHCP, the exclusion was implemented by dividing the dynamic address space into separate pools and intentionally leaving a protected gap between them.

Environment

- **Subnet:** `10.20.20.0/24`
- **DHCP Server / Gateway:** pfSense01 — `10.20.20.254`
- **DNS Server:** DC01 — `10.20.20.10`
- **Original DHCP Pool:** `10.20.20.100 – 10.20.20.200`
- **Test Client:** WIN01
- **WIN01 Address:** `10.20.20.140`
- **Reserved ADMINPC01 Address:** `10.20.20.90`

Step 1 — Review the Existing DHCP Scope

The LAN DHCP configuration was reviewed in pfSense.

The original dynamic allocation pool was:

```
10.20.20.100 – 10.20.20.200
```

Before modifying the pool, the active DHCP leases were inspected to ensure the proposed exclusion would not interfere with an actively leased address.

WIN01 was using `10.20.20.140`, while ADMINPC01 used its previously configured DHCP reservation at `10.20.20.90`.

Step 2 — Define the Excluded Address Range

The following addresses were selected for exclusion:

```
10.20.20.150 – 10.20.20.160
```

The purpose was to prevent DHCP from dynamically allocating these addresses while preserving them for potential infrastructure or manually addressed devices.

Step 3 — Reconfigure the DHCP Address Pools

Unlike Windows Server DHCP, the pfSense ISC DHCP interface used in the lab does not provide the same explicit **Exclusion Range** configuration.

The exclusion concept was therefore implemented by dividing the dynamic address space into two pools:

```
Primary DHCP Pool:
10.20.20.100 – 10.20.20.149

Excluded Gap:
10.20.20.150 – 10.20.20.160

Additional DHCP Pool:
10.20.20.161 – 10.20.20.200
```

This removed `.150–.160` from all dynamic allocation pools.

Step 4 — Save and Apply the Configuration

The updated DHCP pool configuration was saved and the changes were applied in pfSense.

The DHCP lease information subsequently showed the updated primary pool:

```
Pool Start: 10.20.20.100
Pool End:   10.20.20.149
Capacity:   50
```

The additional pool remained configured for `10.20.20.161–200`.

Step 5 — Validate DHCP Client Operation

WIN01 was used to verify that DHCP continued functioning after the pool modification.

The existing lease was released:

```
ipconfig /release
```

A new DHCP lease was then requested:

```
ipconfig /renew
```

WIN01 successfully received:

```
IPv4 Address:    10.20.20.140
Subnet Mask:     255.255.255.0
Default Gateway: 10.20.20.254
DNS Suffix:      benlab.local
```

The assigned address `.140` belongs to the permitted primary pool of `.100–.149` and does not fall within the protected `.150–.160` range.

Validation Result

Task 4 completed successfully. ✅

The DHCP allocation space was reorganized so that:

`.100–.149` → Dynamic DHCP
`.150–.160` → Excluded from dynamic allocation
`.161–.200` → Dynamic DHCP

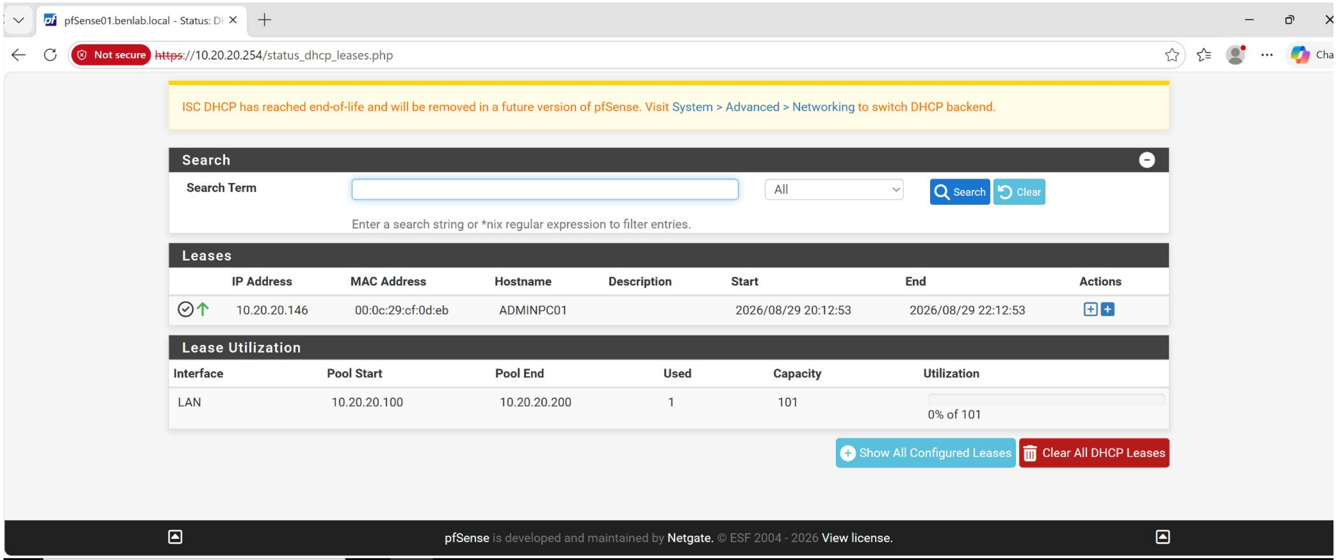
WIN01 successfully renewed its lease after the change, confirming that DHCP service remained operational.

The server-side pool configuration provides the primary evidence that `.150–.160` is outside the dynamic allocation ranges.

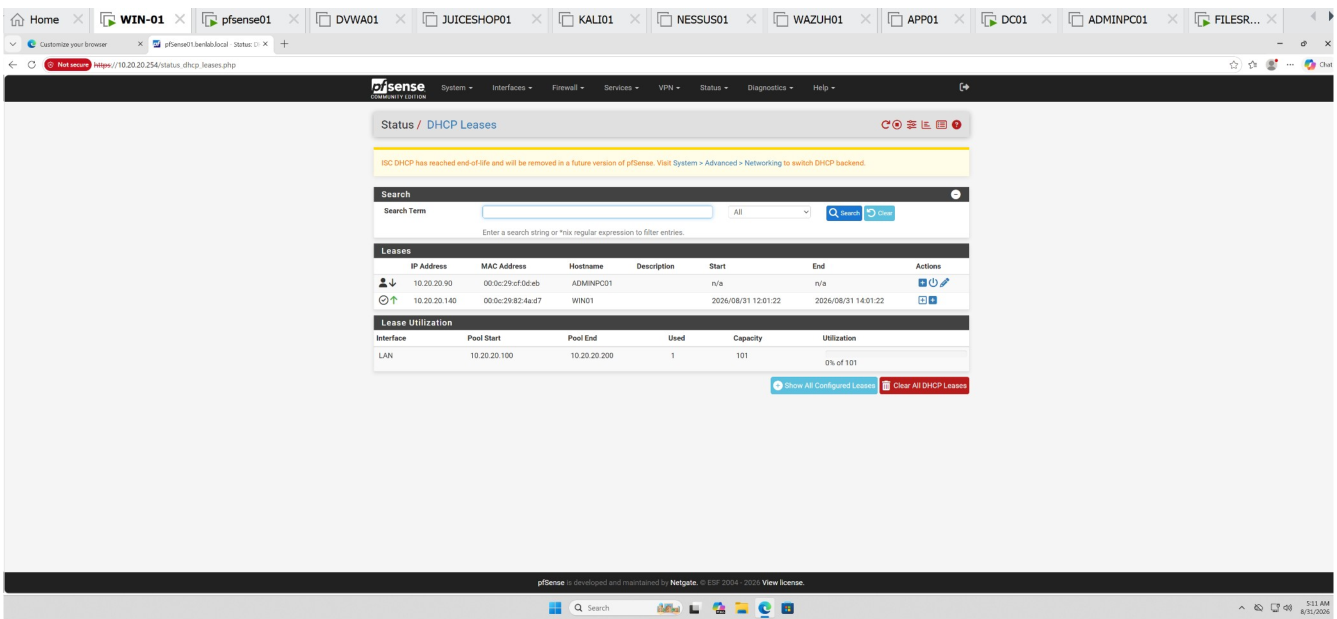
Key Learning Points

- DHCP exclusions prevent selected IP addresses from being dynamically assigned.
- Excluded addresses can be useful for infrastructure, appliances, or other devices requiring separately managed addressing.
- Windows Server DHCP supports explicit exclusion ranges within a scope.
- In this pfSense ISC DHCP implementation, the same allocation-control objective was achieved by creating separate DHCP pools with an intentional address gap.
- Active leases should be inspected **before modifying DHCP ranges** to avoid disrupting clients.
- `ipconfig /release` and `ipconfig /renew` can be used to validate DHCP operation after configuration changes.
- DHCP concepts remain consistent even when different DHCP products implement them differently.

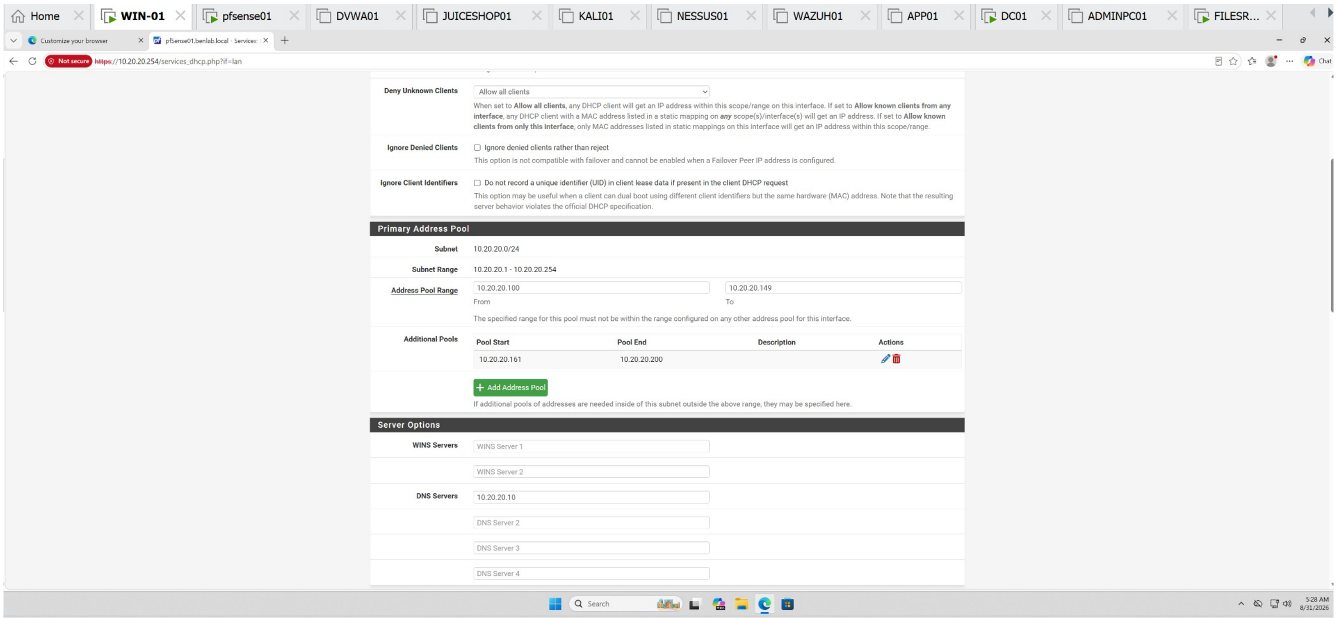
Screenshots / Evidence



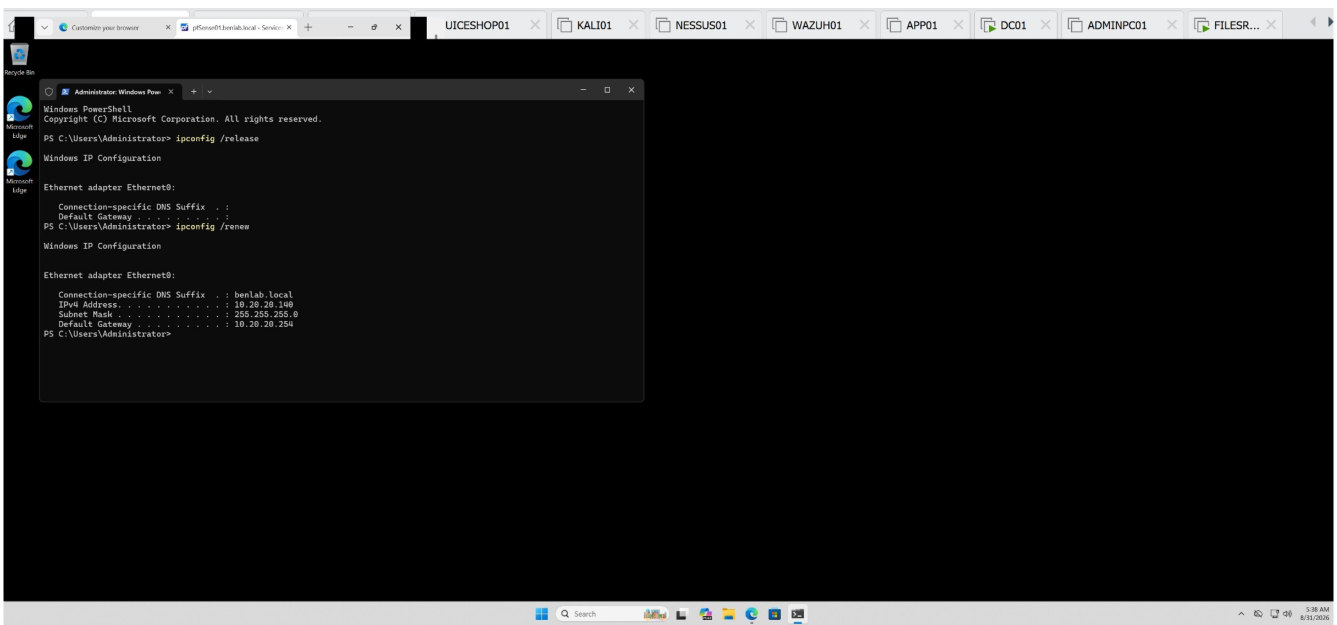
Original pfSense pool showing `10.20.20.100–200`.



DHCP leases before the change showing WIN01 at `.140`.



Final pfSense configuration showing **Primary `.100–149` and Additional `.161–200`**.



WIN01 `/release` + `/renew` showing successful reassignment of `10.20.20.140`.

Task 5 – Configure and Validate DHCP Scope Options

Objective

Configure and validate DHCP scope options on **pfSense01** to ensure DHCP clients receive the correct network configuration for the BENLAB environment.

Environment

DHCP Server: pfSense01
DHCP Server / Gateway IP: `10.20.20.254`
DNS Server: DC01 — `10.20.20.10`
Domain: `benlab.local`
Test Client: WIN01
Network: `10.20.20.0/24`

DHCP Scope Options Reviewed

On **pfSense01**, navigated to:

Services → DHCP Server → LAN

The DHCP configuration was reviewed to verify the network options distributed to DHCP clients.

The following settings were confirmed:

DNS Server: `10.20.20.10`
Default Gateway: `10.20.20.254`
Domain Name: `benlab.local`
Default Lease Time: `7200 seconds`
Maximum Lease Time: `86400 seconds`

The configuration ensures that BENLAB DHCP clients use **DC01 for internal DNS resolution** while using **pfSense01 as their default gateway**.

Client-Side Validation

On **WIN01**, opened an elevated PowerShell session and executed:

```
ipconfig /all
```

The resulting configuration confirmed that WIN01 was using DHCP and had received the expected scope information.

Observed values included:

```
DHCP Enabled:                     Yes
IPv4 Address:                    10.20.20.140
Subnet Mask:                     255.255.255.0
Default Gateway:                 10.20.20.254
DHCP Server:                     10.20.20.254
DNS Servers:                     10.20.20.10
Connection-specific DNS Suffix: benlab.local
```

WIN01 also received an approximately **two-hour DHCP lease**, consistent with the configured default lease time of `7200` seconds.

Validation Result

The DHCP scope options were successfully validated.

pfSense01 correctly provides DHCP configuration to WIN01, including the appropriate default gateway, DNS server, domain suffix, subnet configuration, and lease information.

This confirms that DHCP clients on the BENLAB network are receiving the required network settings automatically and that **pfSense01 DHCP and DC01 DNS are operating together as designed**.

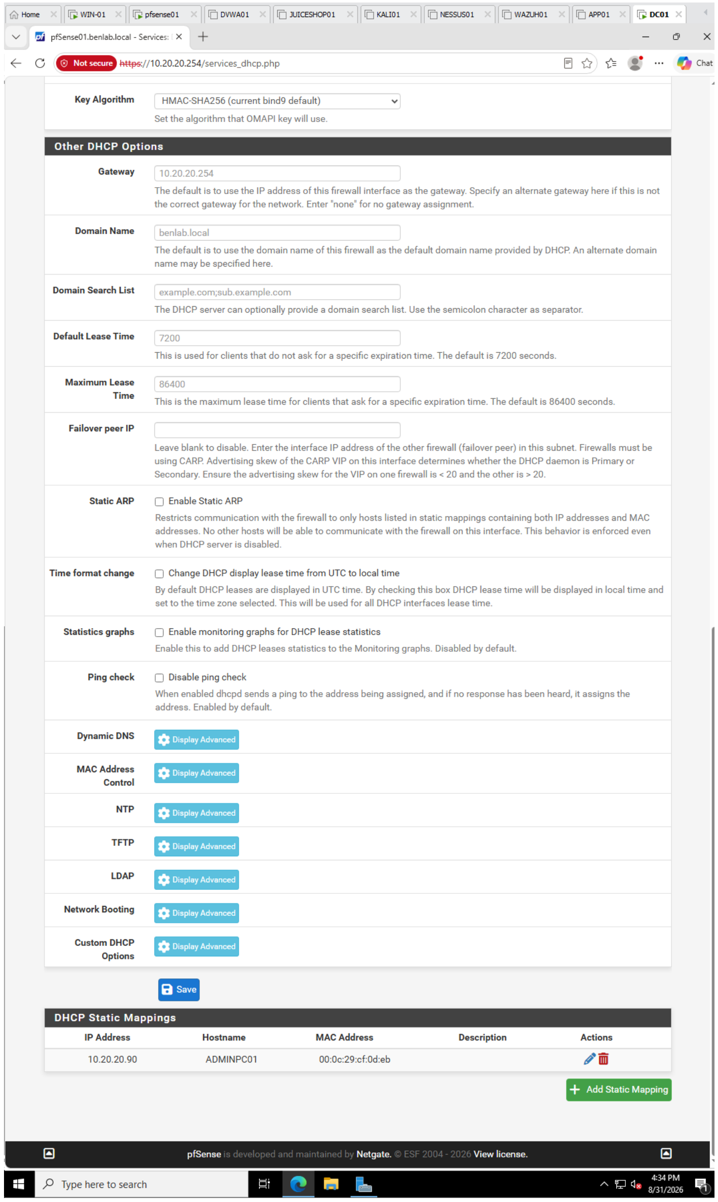
Skills Demonstrated

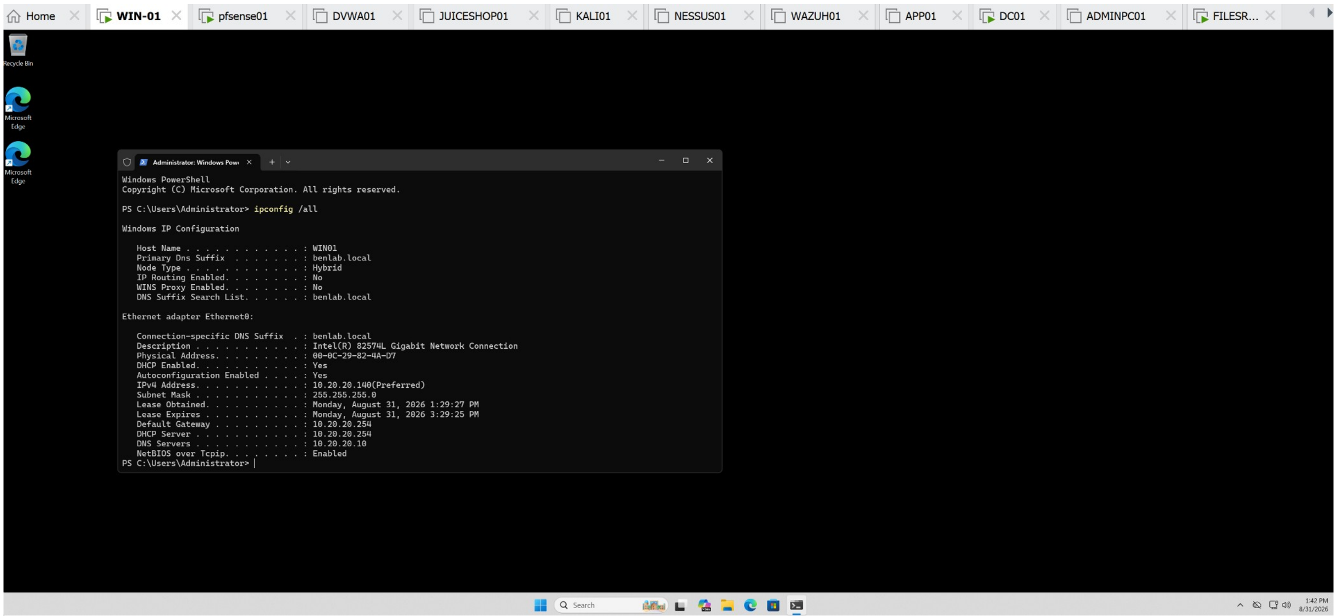
DHCP scope option administration, DHCP client configuration validation, DNS and gateway assignment, lease configuration verification, Windows `ipconfig` troubleshooting, and integration of DHCP with Active Directory DNS.

Screenshots / Evidence

pfSense DHCP Scope Options

Show the pfSense section containing `DNS Server: 10.20.20.10`, `Gateway: 10.20.20.254`, and `Domain Name: benlab.local`.





Screenshot 2 — WIN01 DHCP Validation
Showing `10.20.20.140`, gateway `10.20.20.254`, DHCP server `10.20.20.254`, DNS `10.20.20.10`, and `benlab.local`.

Task 6 — DHCP Troubleshooting & Client Validation

Objective:
Validate DHCP client operation on WIN01 and demonstrate a practical troubleshooting procedure by releasing and renewing its DHCP lease. Confirm that pfSense01 correctly provides IPv4 addressing and gateway information while DC01 provides DNS services.

Environment

- **Client:** WIN01
- **Domain:** `benlab.local`
- **DHCP Server / Gateway:** pfSense01 — `10.20.20.254`
- **DNS Server:** DC01 — `10.20.20.10`
- **Network:** `10.20.20.0/24`

- **DHCP Client Address:** `10.20.20.140`

Step 1 — Validate Existing DHCP Configuration

On WIN01, the current network configuration was reviewed using:

```
ipconfig /all
```

The workstation was confirmed to be configured as a DHCP client and had received a valid address from pfSense01.

Observed configuration:

```
IPv4 Address:     10.20.20.140
Subnet Mask:      255.255.255.0
Default Gateway:  10.20.20.254
DHCP Server:      10.20.20.254
DNS Server:       10.20.20.10
DNS Suffix:       benlab.local
DHCP Enabled:     Yes
```

Result: WIN01 had a valid DHCP configuration before troubleshooting began.

Step 2 — Release the DHCP Lease

The existing DHCP configuration was intentionally released:

```
ipconfig /release
```

After releasing the lease, WIN01 no longer had its assigned IPv4 address or default gateway.

Network connectivity temporarily disappeared as expected.

Purpose: This simulated a client without a usable DHCP lease and demonstrated the effect of removing the workstation's current DHCP configuration.

Step 3 — Renew the DHCP Lease

A new DHCP request was initiated:

```
ipconfig /renew
```

WIN01 successfully contacted the DHCP service on pfSense01 and obtained:

```
IPv4 Address:     10.20.20.140
Subnet Mask:      255.255.255.0
Default Gateway:  10.20.20.254
```

Network connectivity returned immediately after the successful renewal.

Result: DHCP communication between WIN01 and pfSense01 was successfully validated.

Step 4 — Final Client Validation

The complete configuration was checked again:

```
ipconfig /all
```

The final output confirmed:

- DHCP remained enabled.
- WIN01 received `10.20.20.140`.
- pfSense01 (`10.20.20.254`) supplied DHCP and gateway services.
- DC01 (`10.20.20.10`) remained the configured DNS server.
- The `benlab.local` DNS suffix was correctly assigned.
- Valid DHCP lease information was present.

Result: WIN01 successfully recovered its complete network configuration through DHCP.

Troubleshooting Concept — APIPA

During the task, APIPA troubleshooting was also reviewed.

A Windows workstation receiving an address in:

```
169.254.0.0/16
```

typically indicates that the DHCP-enabled client was unable to successfully obtain an IPv4 lease.

The troubleshooting workflow established for the lab was:

Check network/NIC connectivity → verify DHCP is enabled → inspect `ipconfig /all` → release/renew the lease → verify the DHCP server and address pool → determine whether one or multiple clients are affected → validate gateway/IP connectivity → troubleshoot DNS after valid IP connectivity is established.

This helps distinguish a **DHCP/addressing problem** from a **DNS resolution problem**.

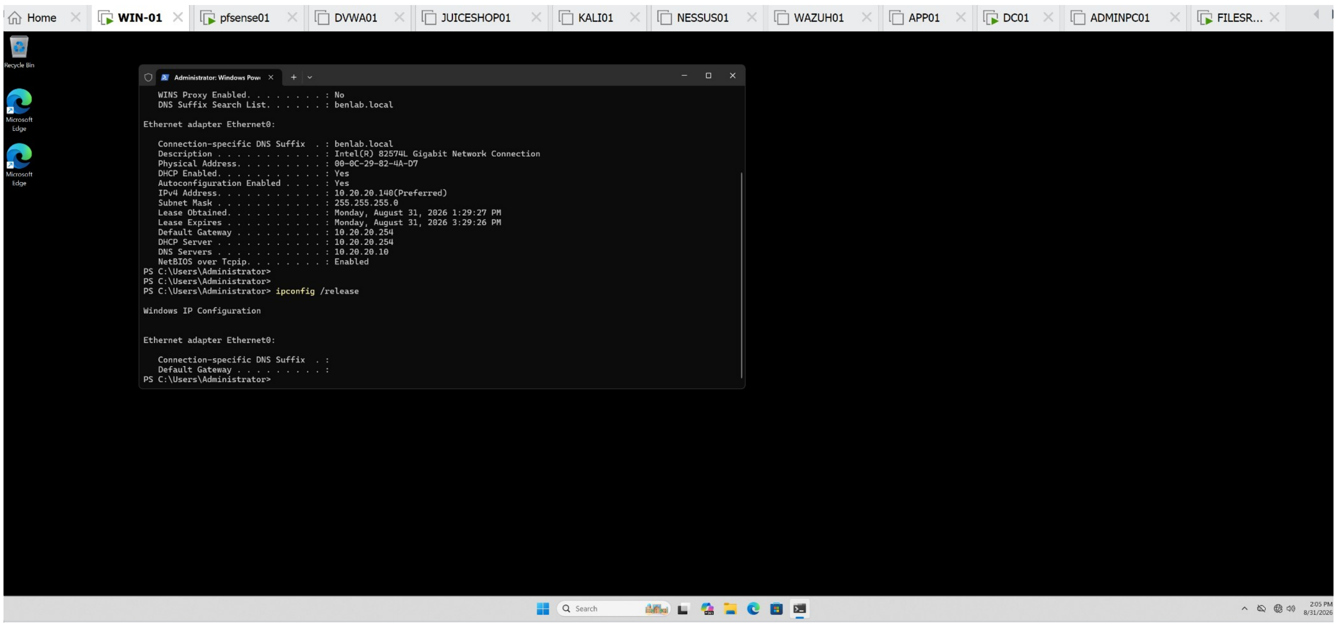
Task Result

Task 6 completed successfully. ✅

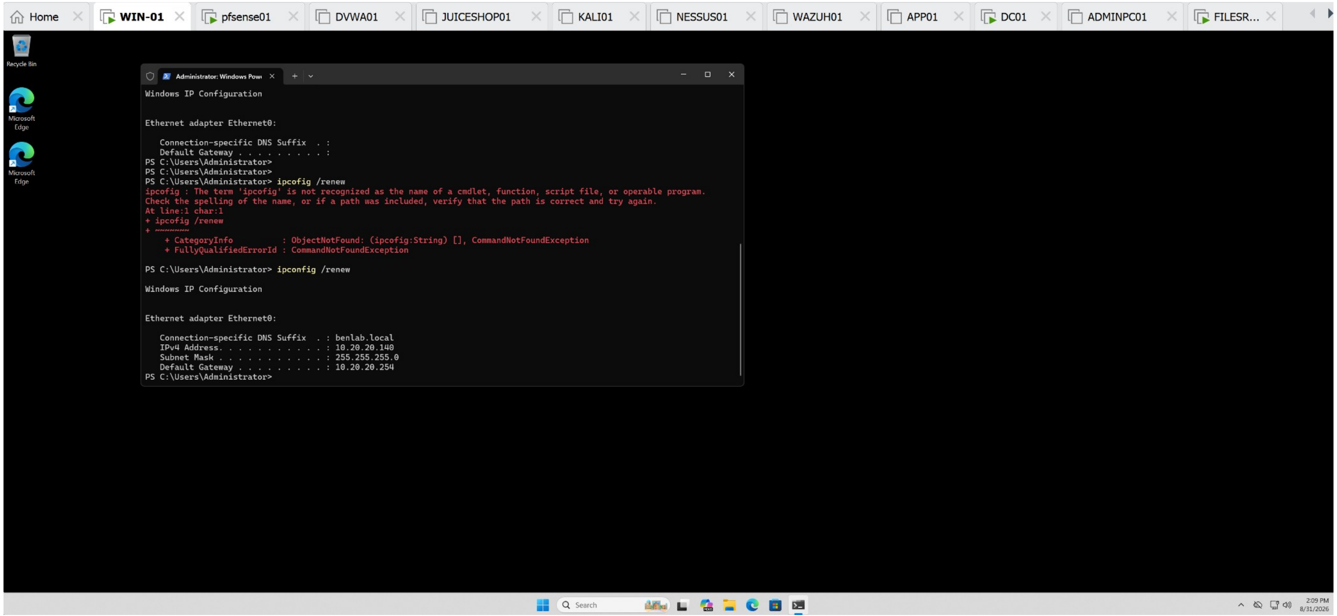
WIN01 demonstrated successful DHCP lease release and renewal. The client recovered its expected `10.20.20.140` address and received the correct gateway and DNS configuration.

The exercise also demonstrated a practical troubleshooting workflow for DHCP failures and APIPA addressing commonly encountered in enterprise workstation support.

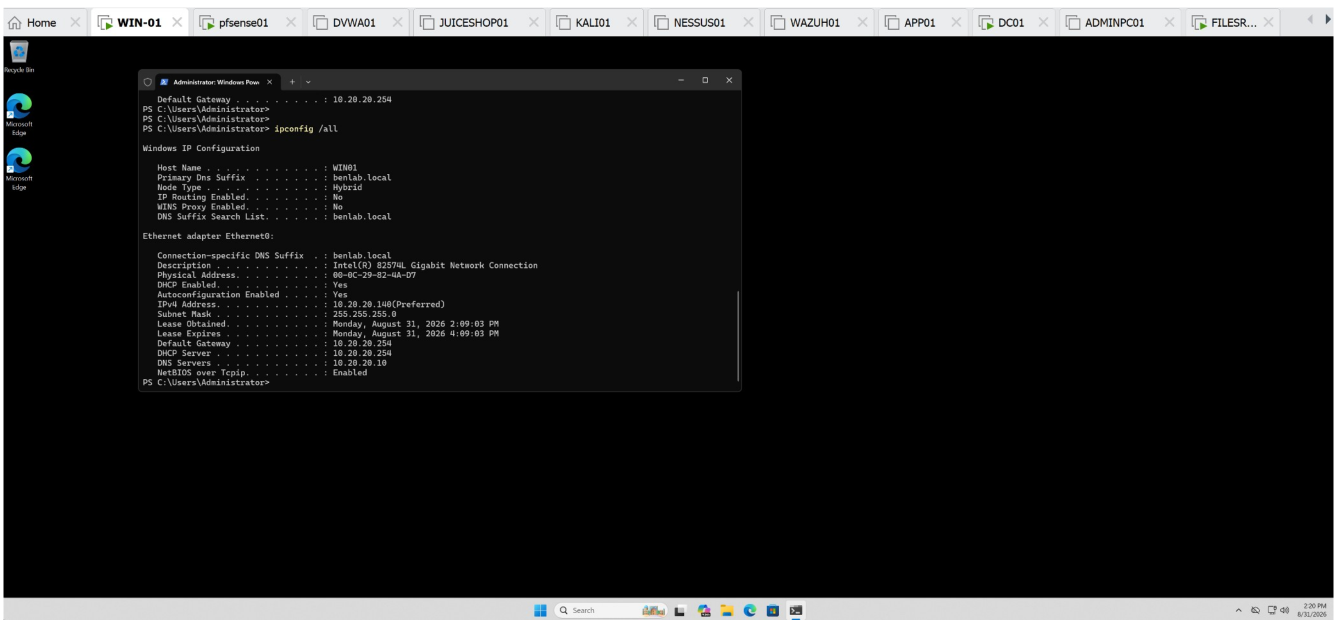
Screenshots / Evidence



WIN01 after `ipconfig /release` — demonstrates loss of the DHCP configuration.



WIN01 after `ipconfig /renew` — demonstrates successful recovery of `10.20.20.140`.



Final `ipconfig /all` — shows DHCP enabled, IPv4 address, gateway, DHCP server, DNS server, and domain suffix together.

Task 7 — DHCP Failure and APIPA Troubleshooting

Objective

Simulate and troubleshoot a DHCP service failure to understand how a Windows workstation behaves when it cannot obtain an IP configuration from the DHCP server.

Procedure

1. Verified WIN01 initially had a valid DHCP configuration.
2. Temporarily disabled the DHCP service on the pfSense LAN interface.
3. Released WIN01's existing DHCP lease:

```
ipconfig /release
```

4. Attempted to obtain a new DHCP lease:

```
ipconfig /renew
```

5. Observed that WIN01 could not contact the DHCP server and the request timed out.
6. Verified that Windows automatically assigned WIN01 an APIPA address:

```
169.254.89.221

Subnet Mask: 255.255.0.0

Default Gateway: None
```

7. Re-enabled DHCP on the pfSense LAN interface.
8. Renewed the DHCP lease again:

```
ipconfig /renew
```

9. Verified that WIN01 successfully recovered its normal DHCP configuration:

```
IPv4 Address:    10.20.20.140

Subnet Mask:     255.255.255.0

Default Gateway: 10.20.20.254
```

10. Tested connectivity to the pfSense gateway:

```
ping 10.20.20.254
```

The gateway responded successfully with **0% packet loss**.

Result

The DHCP failure and recovery process was successfully demonstrated. When the DHCP server became unavailable, WIN01 was unable to renew its lease and automatically assigned itself an APIPA address from the `169.254.0.0/16` range.

After DHCP service was restored, WIN01 successfully obtained its normal `10.20.20.140/24` configuration and regained connectivity to the default gateway.

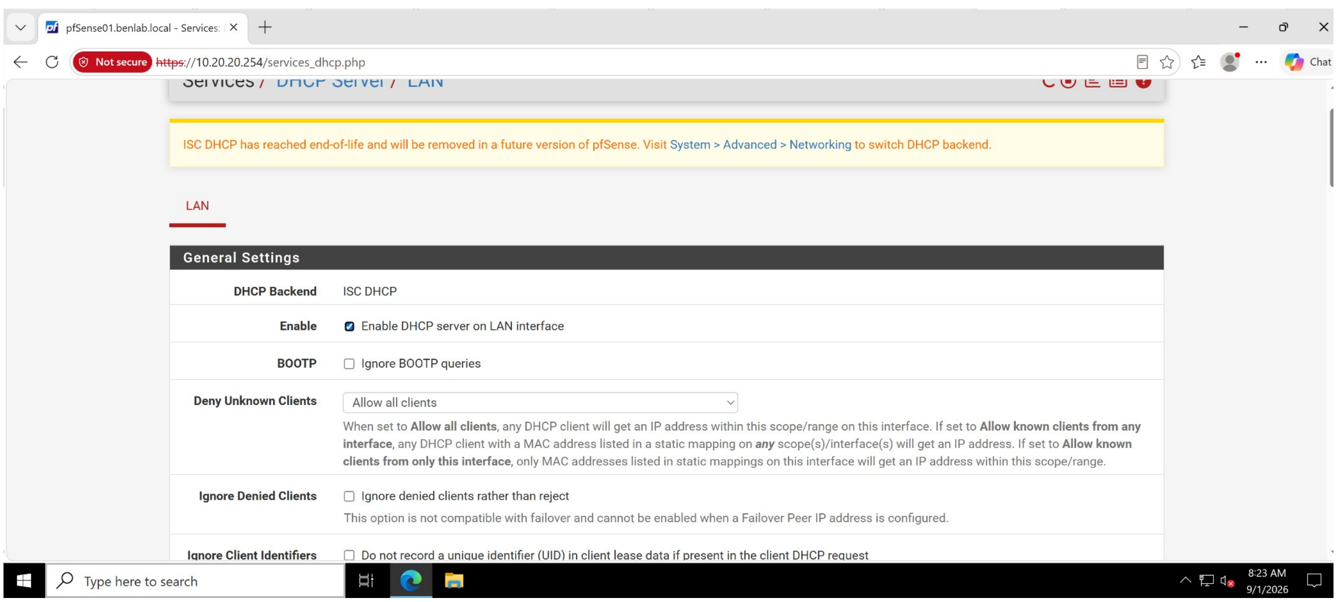
Key Learning

An APIPA (`169.254.x.x`) address is an important troubleshooting indicator on a DHCP-enabled Windows workstation. It indicates that the workstation was unable to obtain a valid IPv4 configuration from a DHCP server.

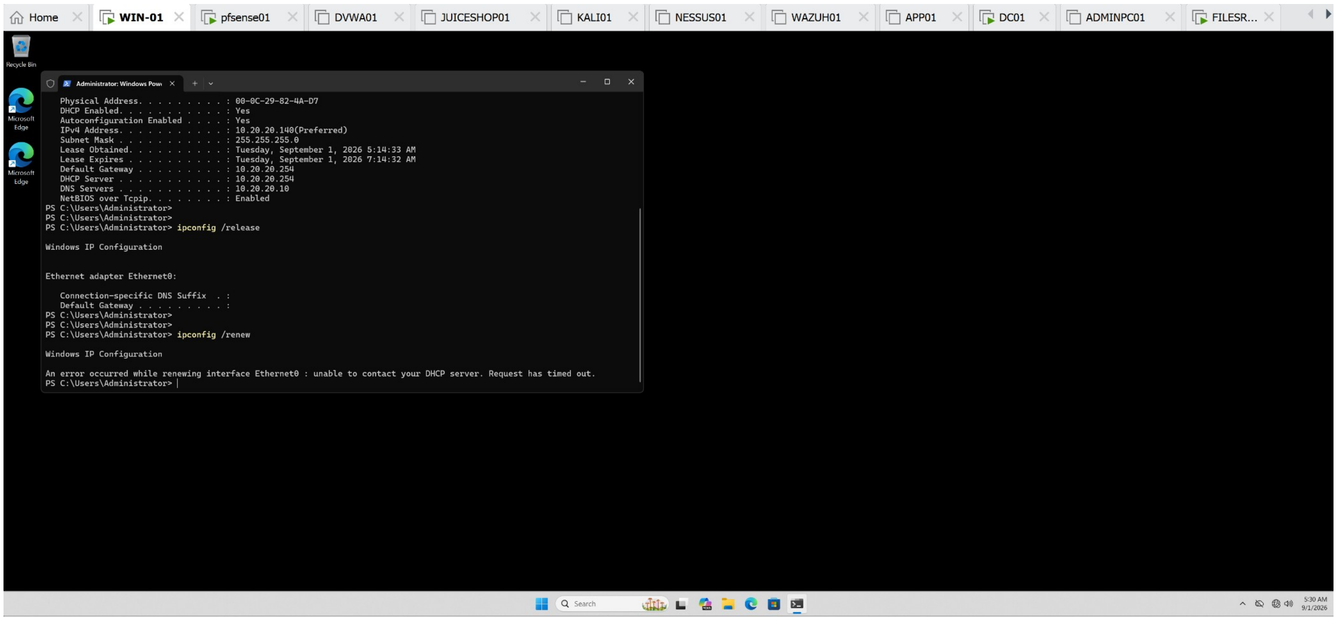
The troubleshooting process demonstrated the relationship:

DHCP unavailable → DHCP request fails → APIPA assigned → DHCP restored → lease renewed → connectivity restored.

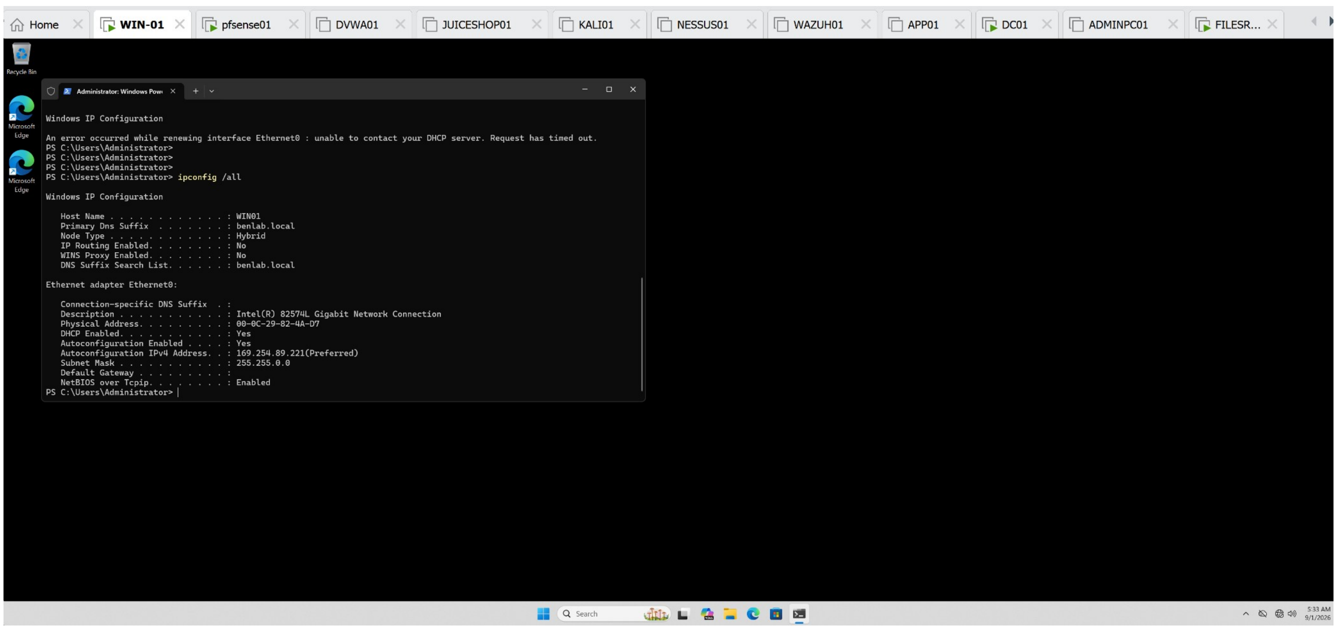
Screenshots / Evidence



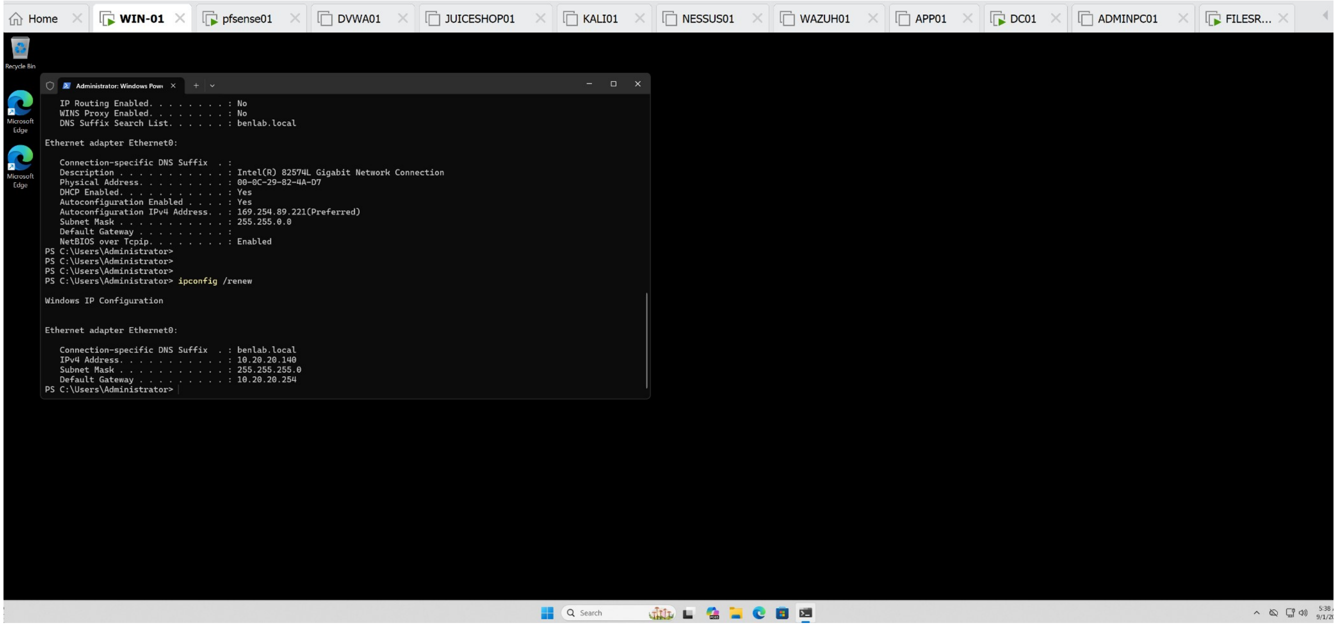
pfSense DHCP service enabled and normal DHCP configuration.



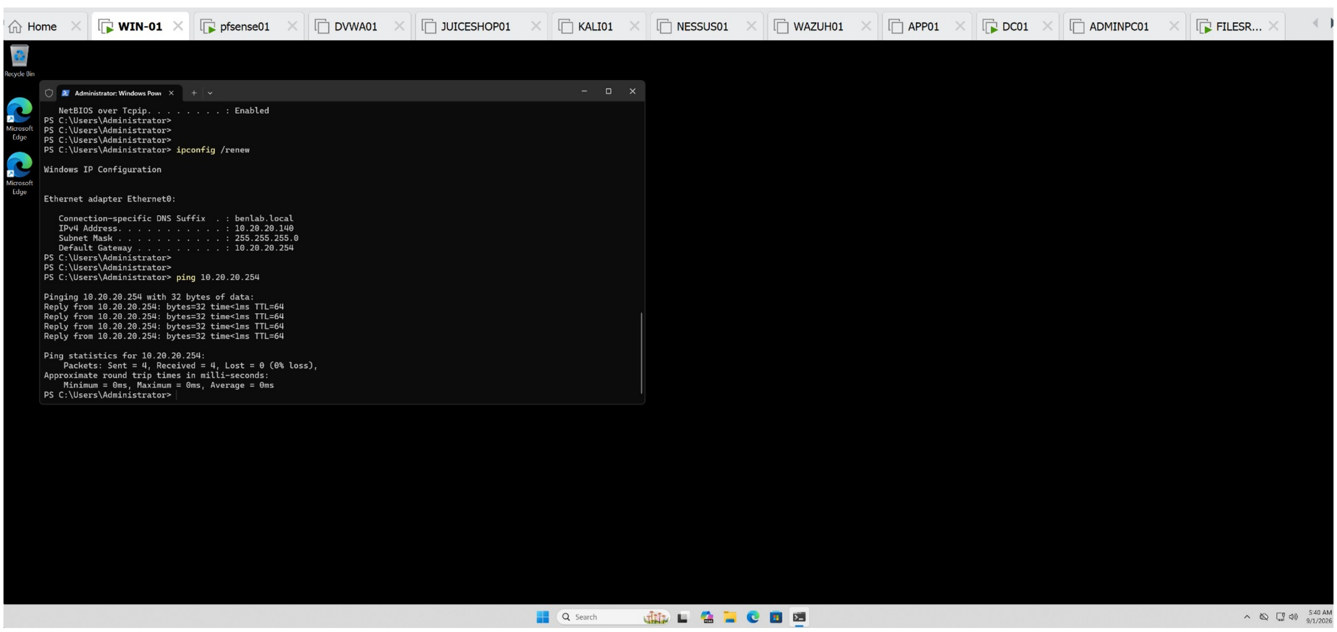
WIN01 DHCP renewal failure after DHCP service was disabled.



`ipconfig /all` showing WIN01 assigned APIPA address `169.254.89.221`.



Screenshot 4: DHCP service restored and `ipconfig /renew` successfully returning WIN01 to `10.20.20.140`.



Screenshot 5: Successful `ping 10.20.20.254` confirming connectivity to the pfSense default gateway after recovery.

Task 8 — DHCP Lease Management

Objective

Manage and validate DHCP client leases using pfSense and a Windows workstation. This task demonstrates the relationship between the DHCP server's lease database and the DHCP configuration maintained locally by a client.

Environment

DHCP Server: pfSense01
DHCP Server / Gateway: `10.20.20.254`
DHCP Scope: `10.20.20.100 – 10.20.20.149`
Client: WIN01
WIN01 MAC Address: `00-0C-29-82-4A-D7`
WIN01 DHCP Address: `10.20.20.140`
DNS Server: DC01 — `10.20.20.10`

Procedure

1. Reviewed the DHCP lease table

Navigated to:

pfSense → Status → DHCP Leases

Verified that WIN01 had an active dynamic DHCP lease for `10.20.20.140`.

The DHCP pool showed one dynamically assigned address in use.

---

2. Verified the client-side DHCP configuration

On WIN01, the following command was used:

```
ipconfig /all
```

The workstation showed:

```
DHCP Enabled:       Yes
IPv4 Address:       10.20.20.140
Subnet Mask:        255.255.255.0
Default Gateway:    10.20.20.254
DHCP Server:        10.20.20.254
DNS Server:         10.20.20.10
```

This established the client's DHCP configuration before manipulating the lease.

---

3. Deleted the WIN01 lease from pfSense

The active `10.20.20.140` lease was manually deleted from the pfSense DHCP lease table.

After deletion, WIN01 disappeared from the active lease list and pfSense reported that no dynamic leases were in use.

---

4. Verified client behavior after server-side deletion

The following command was run again on WIN01:

```
ipconfig /all
```

WIN01 continued using `10.20.20.140` even though its corresponding lease entry had been removed from pfSense.

This demonstrated that **removing a DHCP lease from the server does not immediately remove the IP configuration already held by the client**.

---

5. Released the client's DHCP configuration

On WIN01:

```
ipconfig /release
```

The workstation relinquished its DHCP configuration. Its IPv4 address and default gateway were removed, temporarily interrupting network connectivity.

---

6. Requested a new DHCP lease

WIN01 was instructed to obtain DHCP configuration again:

```
ipconfig /renew
```

pfSense successfully responded to the DHCP request and assigned:

```
IPv4 Address:       10.20.20.140
Subnet Mask:        255.255.255.0
Default Gateway:    10.20.20.254
```

Receiving the same address was valid because `10.20.20.140` was available for assignment again.

---

7. Verified the newly created lease on pfSense

Returned to:

pfSense → Status → DHCP Leases

WIN01 appeared again with:

```
IP Address:    10.20.20.140
MAC Address:   00:0c:29:82:4a:d7
Hostname:      WIN01
Start:         2026/09/01 13:55:42
End:           2026/09/01 15:55:42
```

The new Start and End timestamps confirmed that pfSense had recorded a new DHCP lease following the client renewal.

Validation

The following were successfully validated:

- DHCP leases can be viewed and managed through pfSense.
- WIN01 successfully obtained addressing from pfSense.
- A DHCP server-side lease entry can be manually removed.
- Removing the server's lease record does not immediately remove the client's current IP configuration.
- `ipconfig /release` relinquishes the Windows client's DHCP configuration.
- `ipconfig /renew` initiates a new DHCP lease request.
- pfSense successfully issued a new lease to WIN01.
- The newly issued lease was recorded in the pfSense DHCP lease table with updated timestamps.

Key Learning

A **DHCP server lease** and a **client's locally held IP configuration** are related but are not the same thing.

Deleting a lease from the DHCP server removes the server's lease record, but the client may continue using its existing address until it releases the configuration, the lease expires, or another DHCP event causes renegotiation.

The Windows commands:

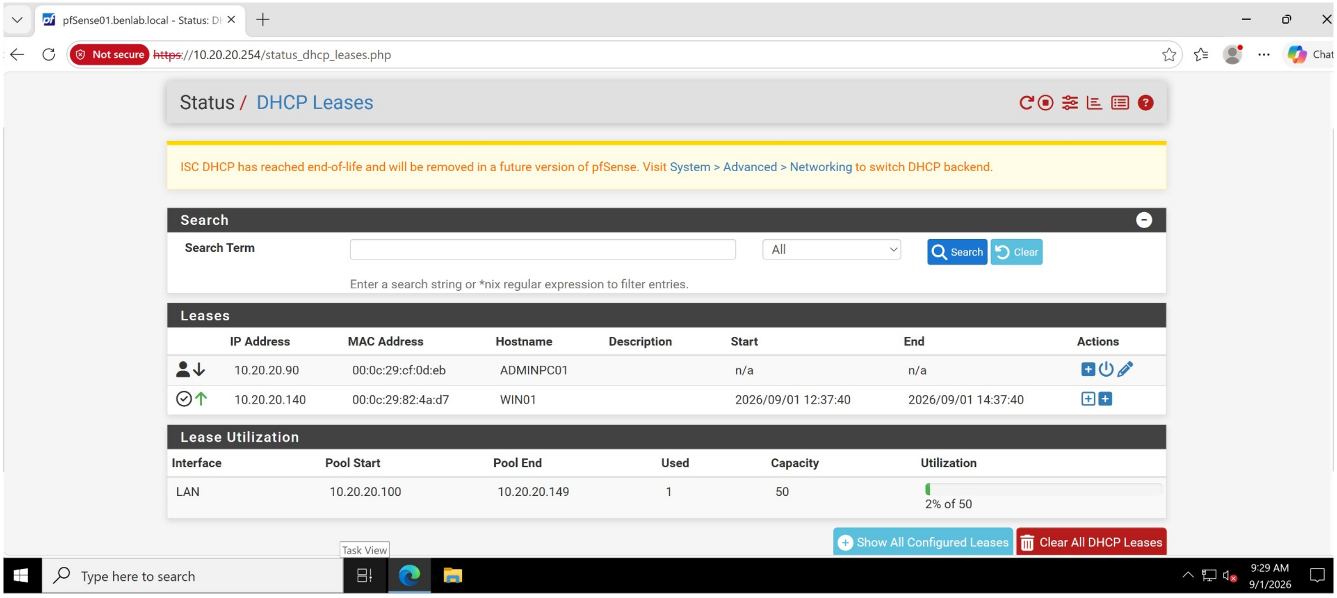
```
ipconfig /release
ipconfig /renew
```

provide direct client-side control over that DHCP lease process.

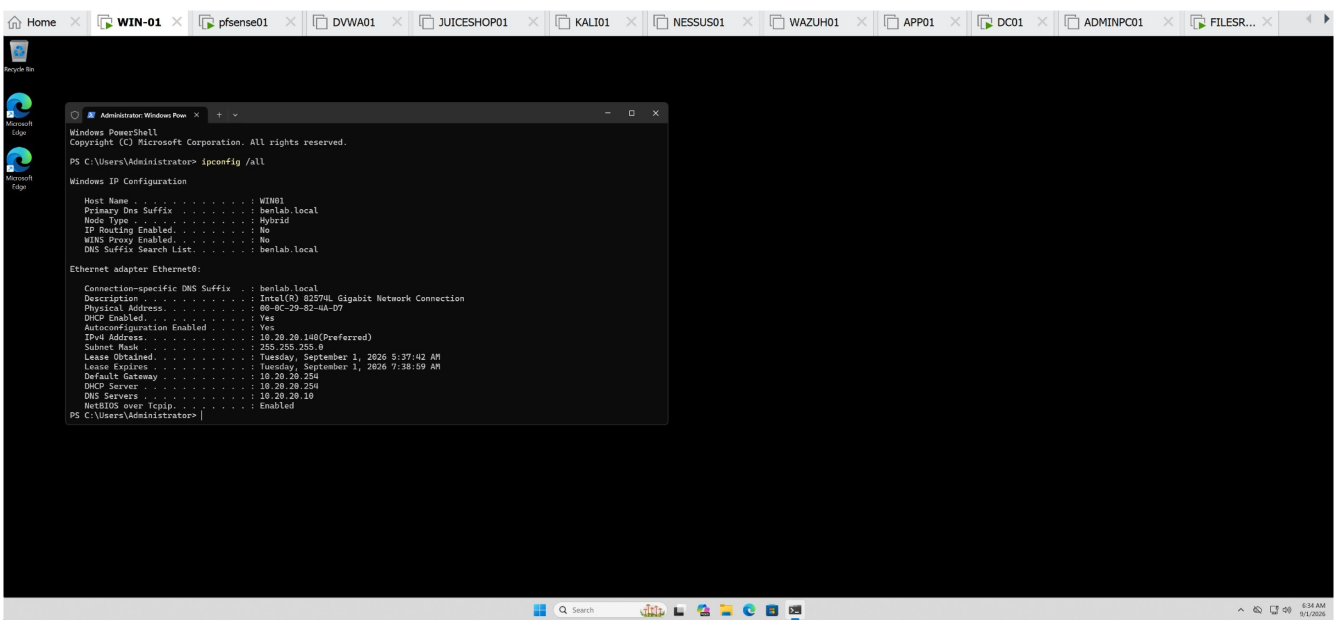
This exercise demonstrated a fundamental DHCP administration workflow:

Inspect → Remove → Observe → Release → Renew → Verify

Screenshots / Evidence

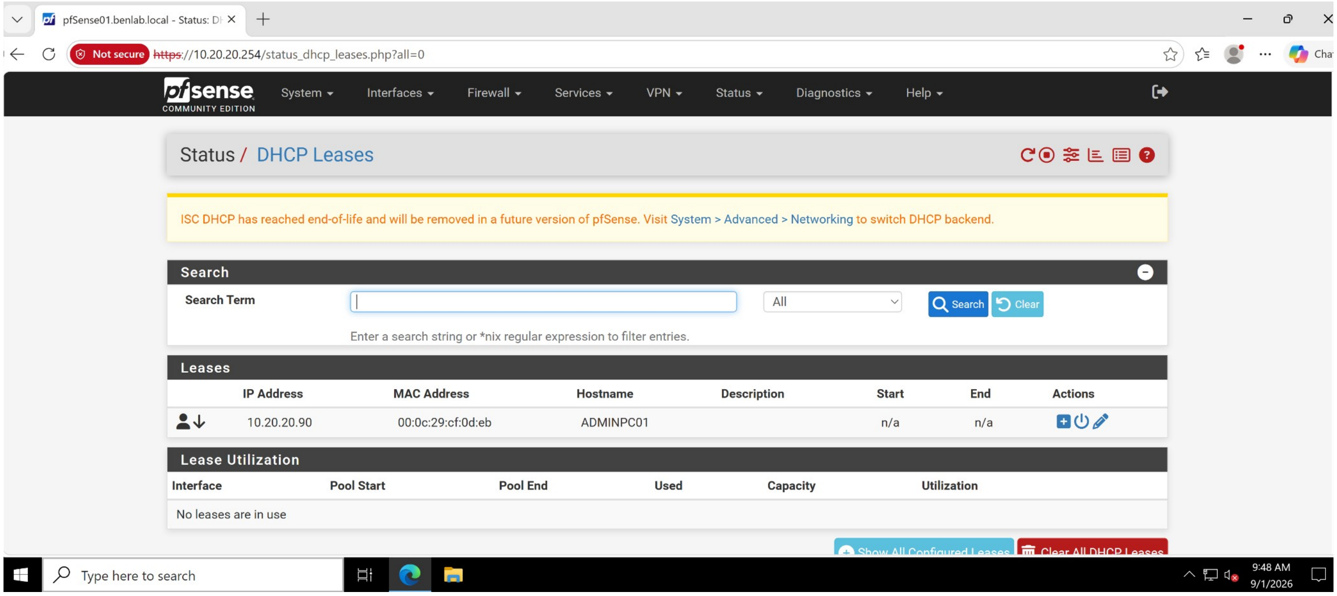


pfSense DHCP Leases showing WIN01 with active address `10.20.20.140`.



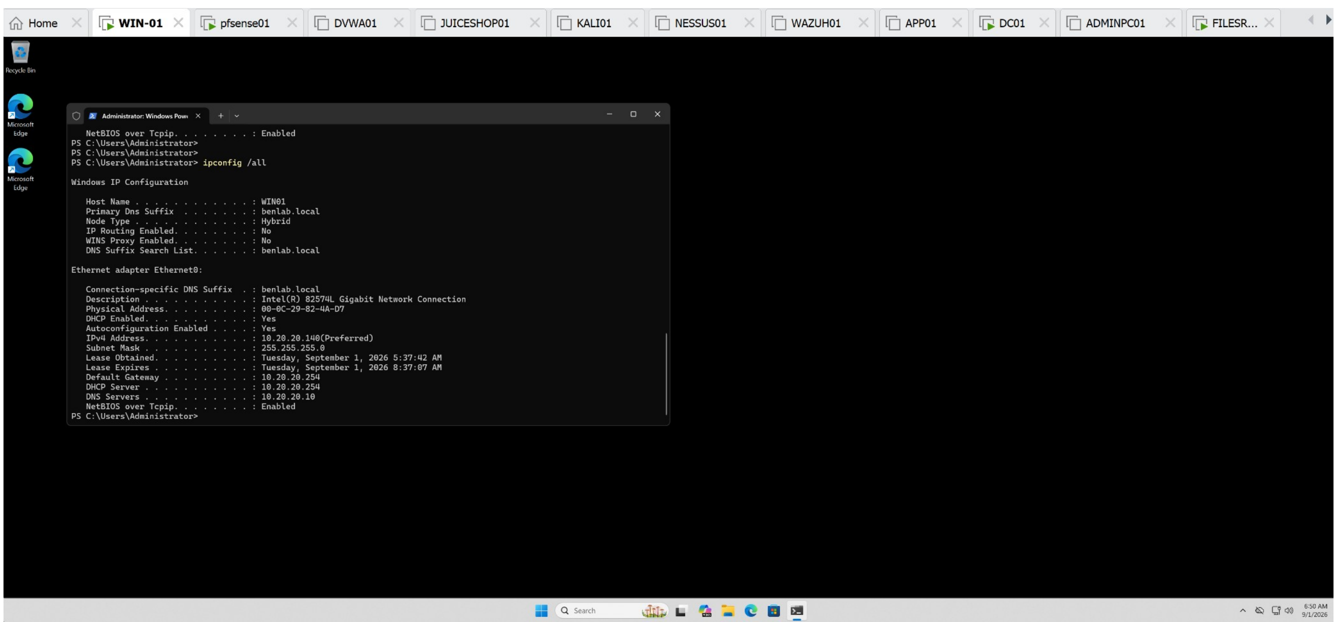
Initial WIN01 DHCP Configuration

`ipconfig /all` showing WIN01's DHCP-assigned address, gateway, DHCP server, DNS server, and lease information.



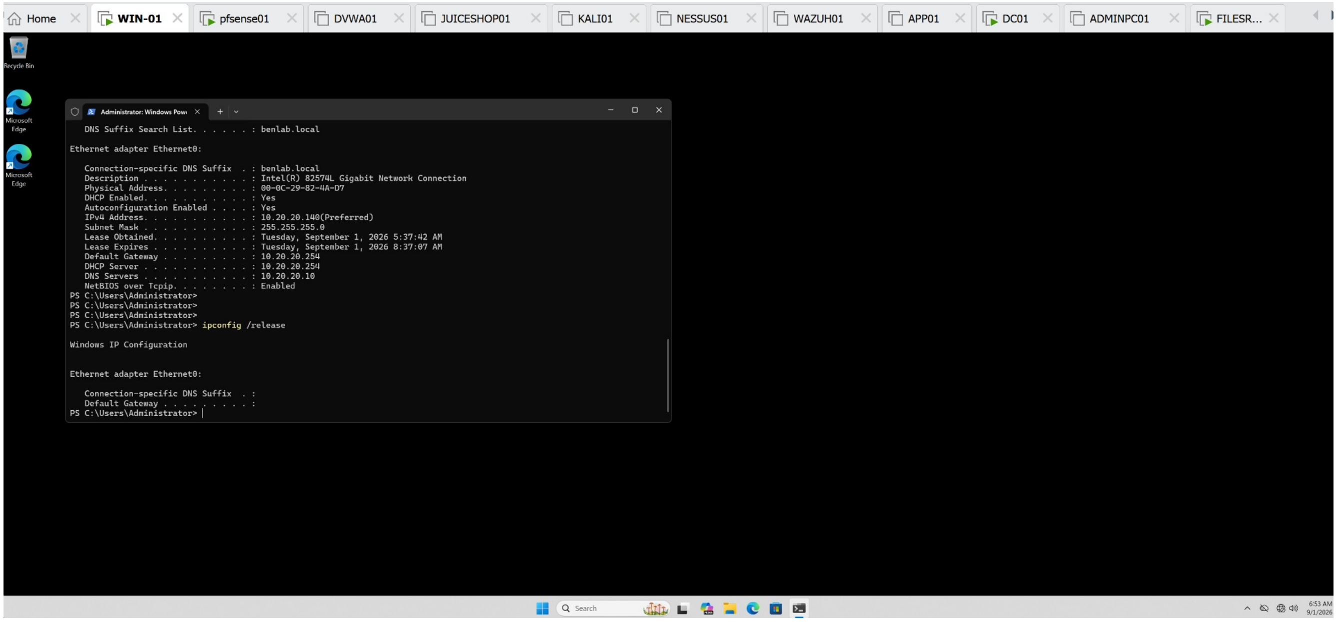
Lease Removed from pfSense

DHCP Leases showing the WIN01 dynamic lease removed and no dynamic leases in use.



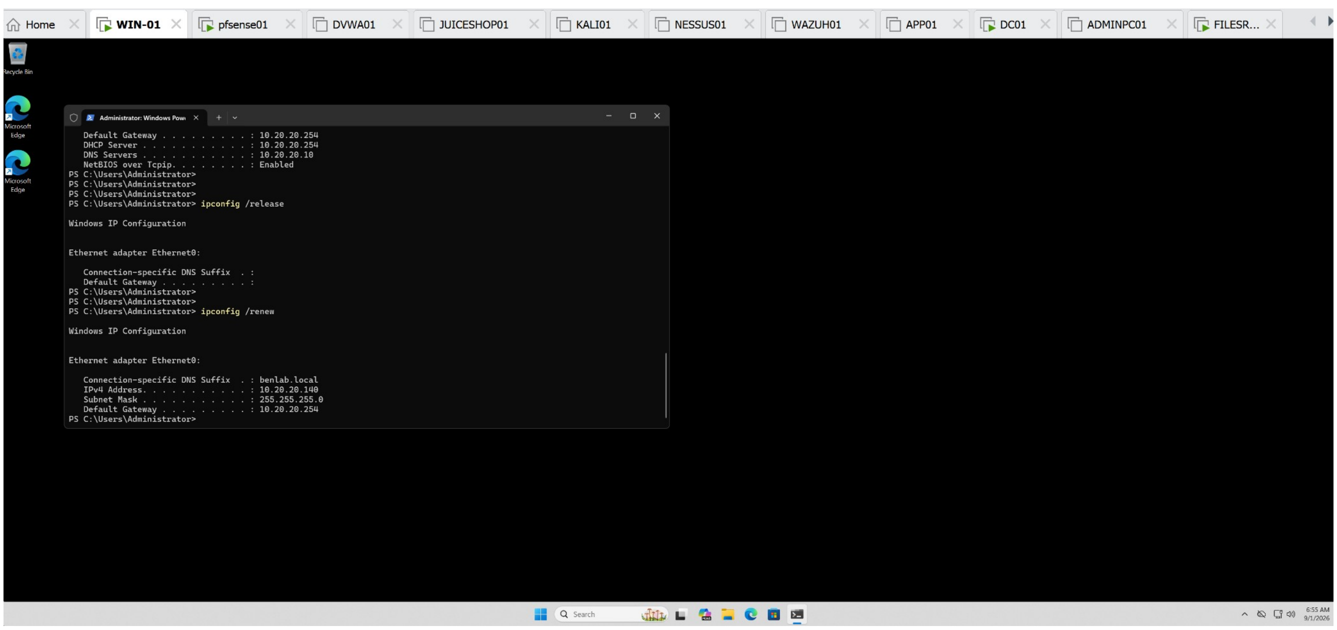
Client Retains Address After Server-Side Deletion

WIN01 `ipconfig /all` showing that `10.20.20.140` remained locally configured after its server-side lease entry was deleted.



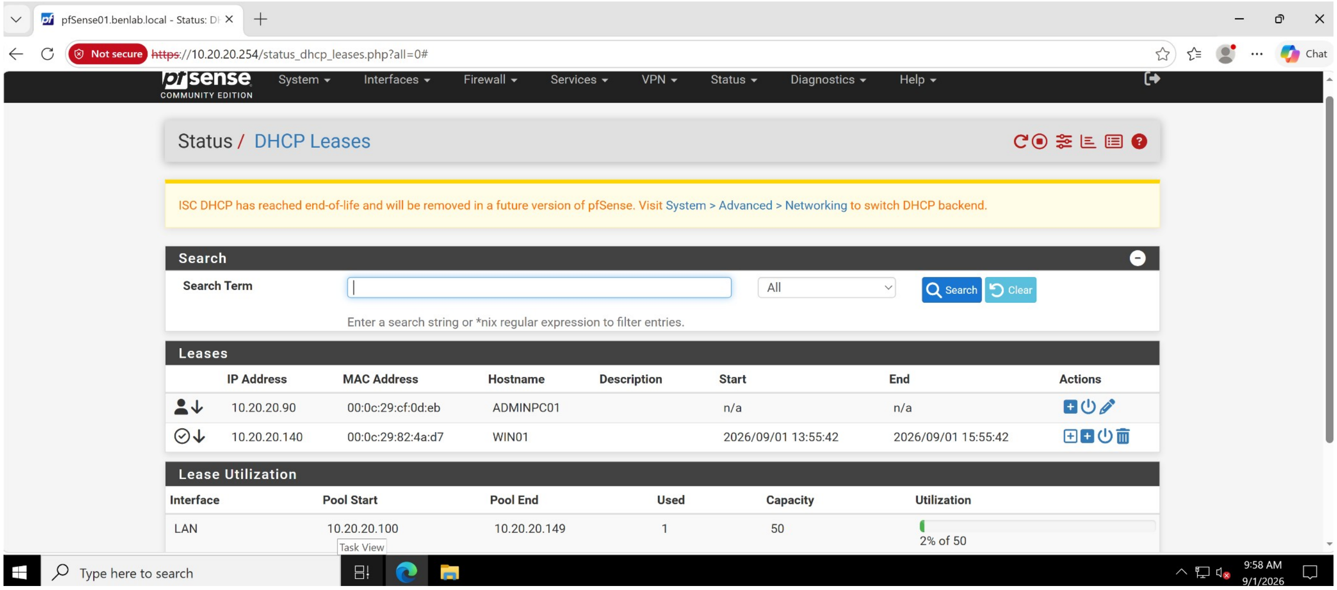
DHCP Release

`ipconfig /release` showing removal of WIN01's DHCP configuration.



DHCP Renewal

`ipconfig /renew` showing WIN01 successfully receiving `10.20.20.140` again.



New Lease Verified in pfSense
pfSense DHCP Leases showing WIN01 at `10.20.20.140` with the new lease timestamps of `13:55:42 – 15:55:42`.

Task 9 — DHCP Service Failure Troubleshooting

Objective

Simulate a DHCP service outage, identify the resulting client-side symptoms, restore DHCP service, and verify successful network recovery.

Environment

DHCP Server / Gateway: pfSense01
LAN Gateway: `10.20.20.254`
DHCP Client: WIN01
Domain: `benlab.local`
Expected WIN01 Address: `10.20.20.140`

Procedure

1. Established the working baseline

WIN01 was initially operating with a valid DHCP configuration:

- IPv4: `10.20.20.140`
- Subnet Mask: `255.255.255.0`
- Default Gateway: `10.20.20.254`
- DNS Server: `10.20.20.10`

---

2. Simulated a DHCP service outage

On pfSense01:

`Services → DHCP Server → LAN`

The **Enable DHCP server on LAN interface** option was disabled and the configuration was saved.

This intentionally prevented pfSense from responding to DHCP requests on the LAN.

---

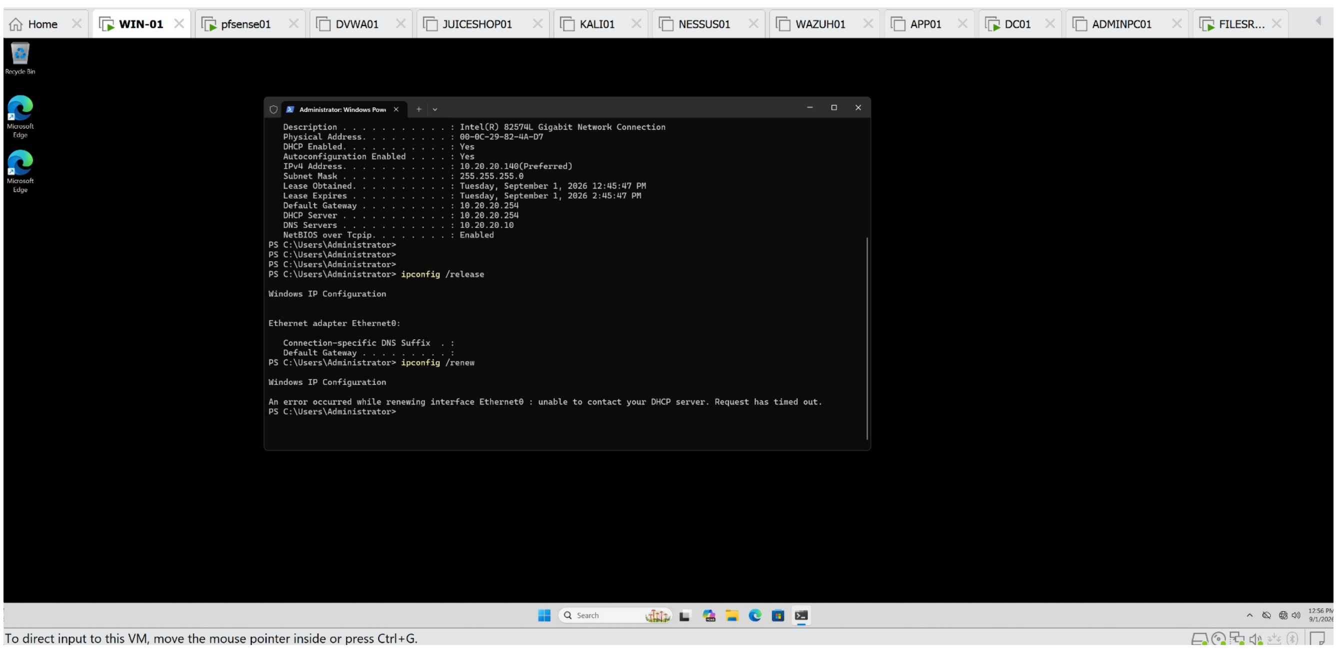
3. Reproduced the client-side failure

On WIN01, the existing DHCP lease was released:

```
ipconfig /release
```

WIN01 then attempted to obtain a new DHCP lease:

```
ipconfig /renew
```



The request failed with:

```
unable to contact your DHCP server. Request has timed out.
```

This confirmed that the client could not obtain network configuration while the DHCP service was unavailable.

---

4. Restored DHCP service

On pfSense01:

`Services → DHCP Server → LAN`

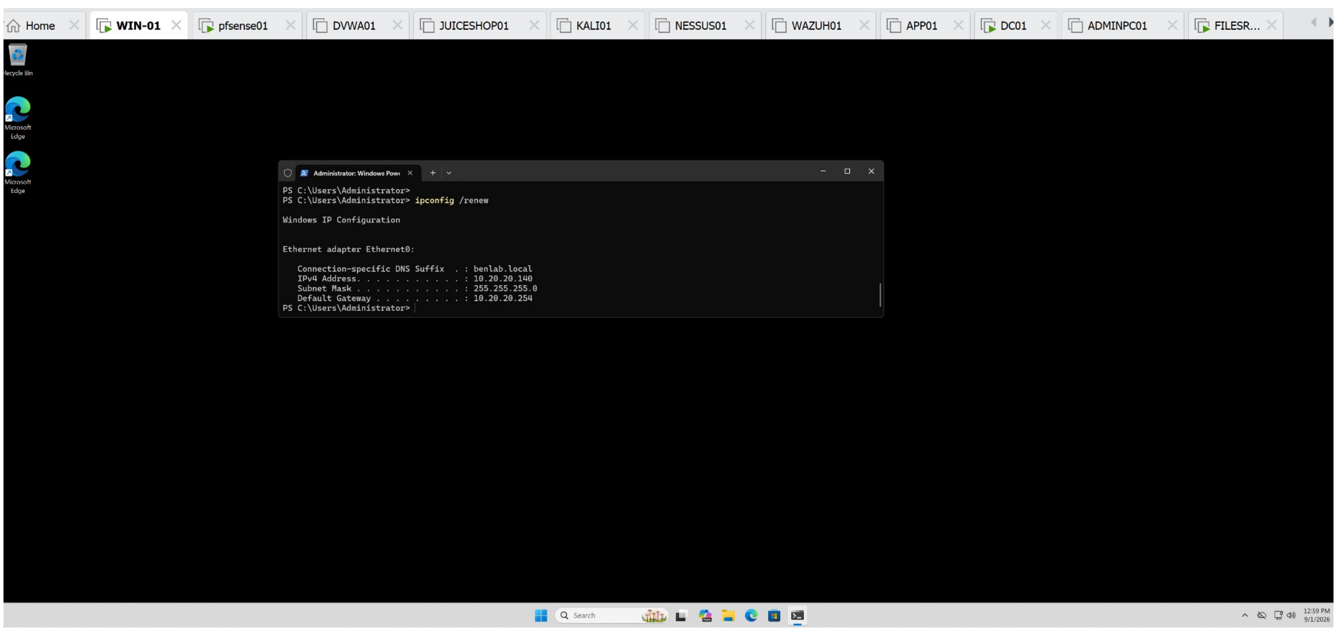
The **Enable DHCP server on LAN interface** option was re-enabled and the configuration was saved.

---

5. Verified DHCP recovery

WIN01 requested a new lease:

```
ipconfig /renew
```



The request succeeded and WIN01 received:

```
IPv4 Address:    10.20.20.140
Subnet Mask:     255.255.255.0
Default Gateway: 10.20.20.254
```

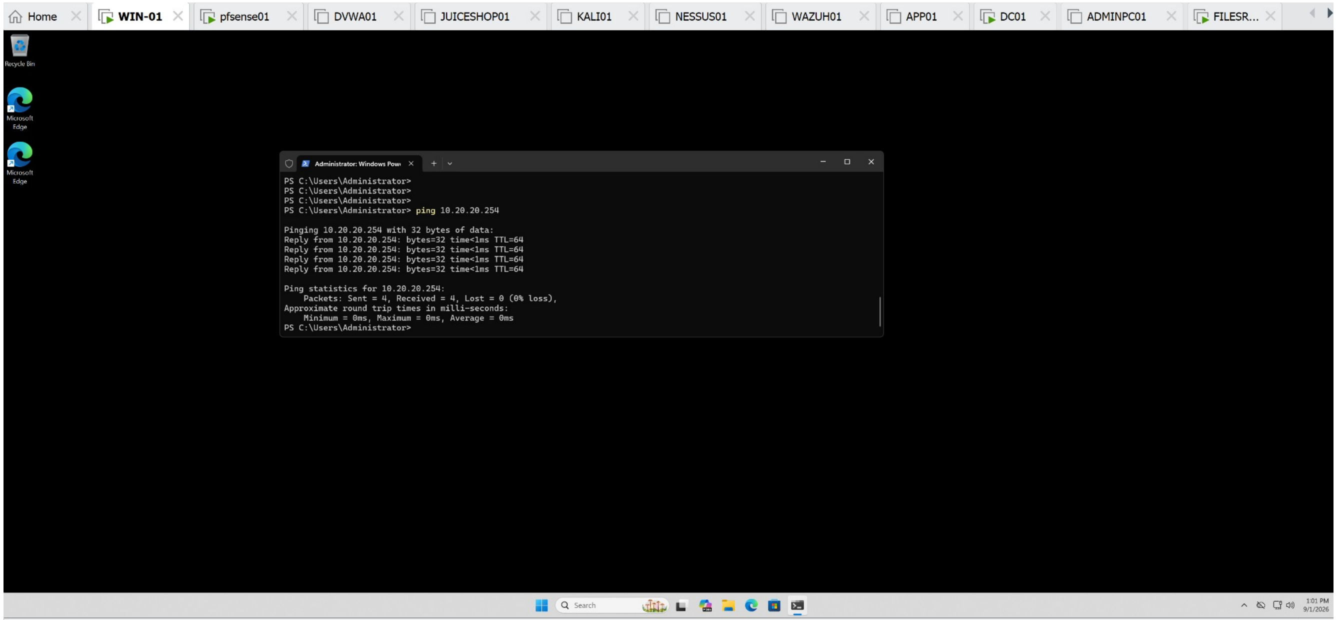
This confirmed successful DHCP lease acquisition after service restoration.

---

6. Verified network connectivity

Connectivity to the default gateway was tested:

```
ping 10.20.20.254
```



Result:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

This verified that WIN01 had regained functional LAN connectivity after DHCP recovery.

Result

A controlled DHCP outage was created by disabling the pfSense LAN DHCP service. WIN01 subsequently failed to obtain a DHCP lease and returned a DHCP server timeout. After restoring the DHCP service, WIN01 successfully reacquired `10.20.20.140`, and connectivity to the `10.20.20.254` gateway was verified with **0% packet loss**.

Skills Demonstrated

`DHCP Troubleshooting | pfSense Administration | DHCP Lease Management | ipconfig | TCP/IP Troubleshooting | Connectivity Validation | Root-Cause Isolation`

Task 10 — DHCP End-to-End Validation & Troubleshooting

Objective:
Validate end-to-end DHCP client configuration, internal DNS resolution, gateway connectivity, external routing, and external DNS resolution from a domain-joined Windows workstation.

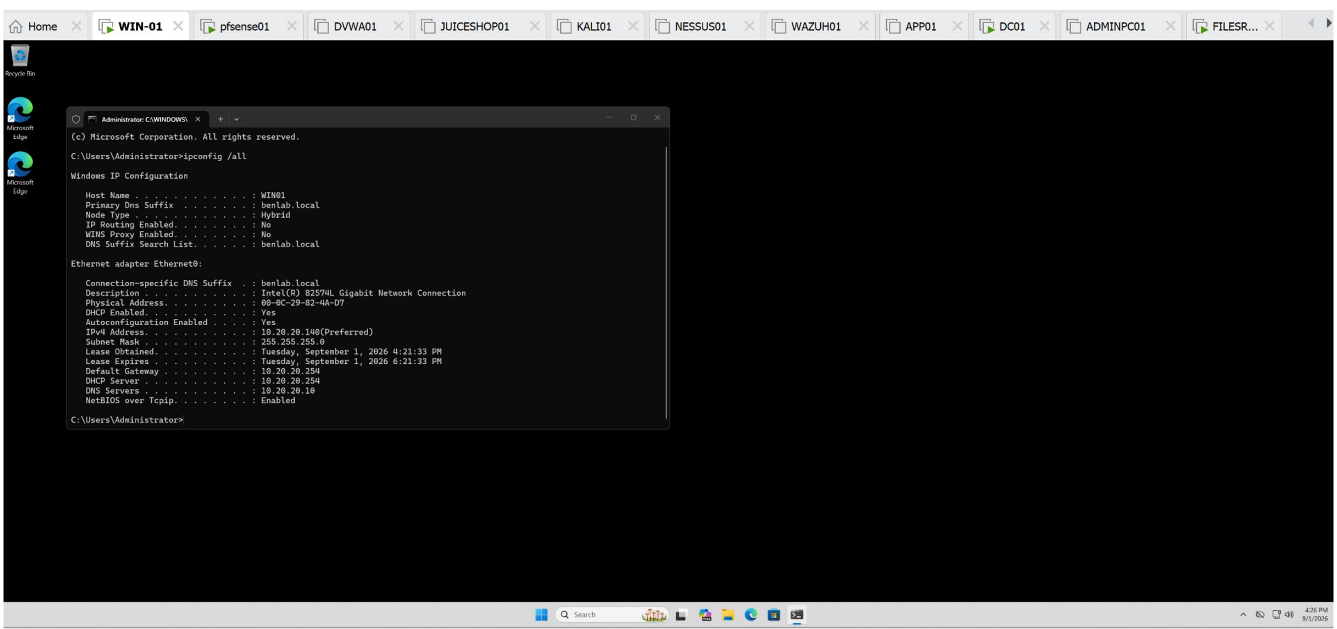
Environment:

- Client: WIN01
- DHCP Server/Gateway: pfSense01 — `10.20.20.254`
- DNS Server/Domain Controller: DC01 — `10.20.20.10`
- Domain: `benlab.local`
- WIN01 DHCP Address: `10.20.20.140`

Validation Performed:

`ipconfig /all` was used to verify that WIN01 received the expected DHCP configuration:

- IPv4 Address: `10.20.20.140`
- Subnet Mask: `255.255.255.0`
- Default Gateway: `10.20.20.254`
- DHCP Server: `10.20.20.254`
- DNS Server: `10.20.20.10`
- DNS Suffix: `benlab.local`



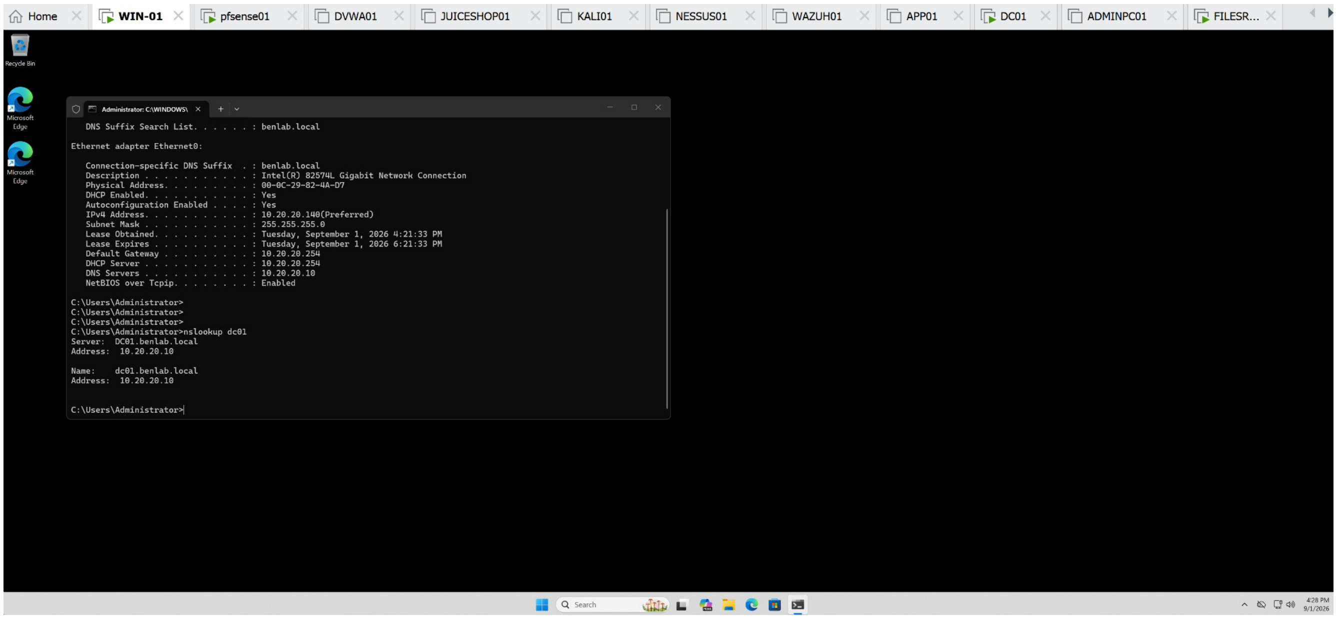
Internal DNS resolution was tested with:

`nslookup dc01`

DC01 successfully resolved to:

`dc01.benlab.local`

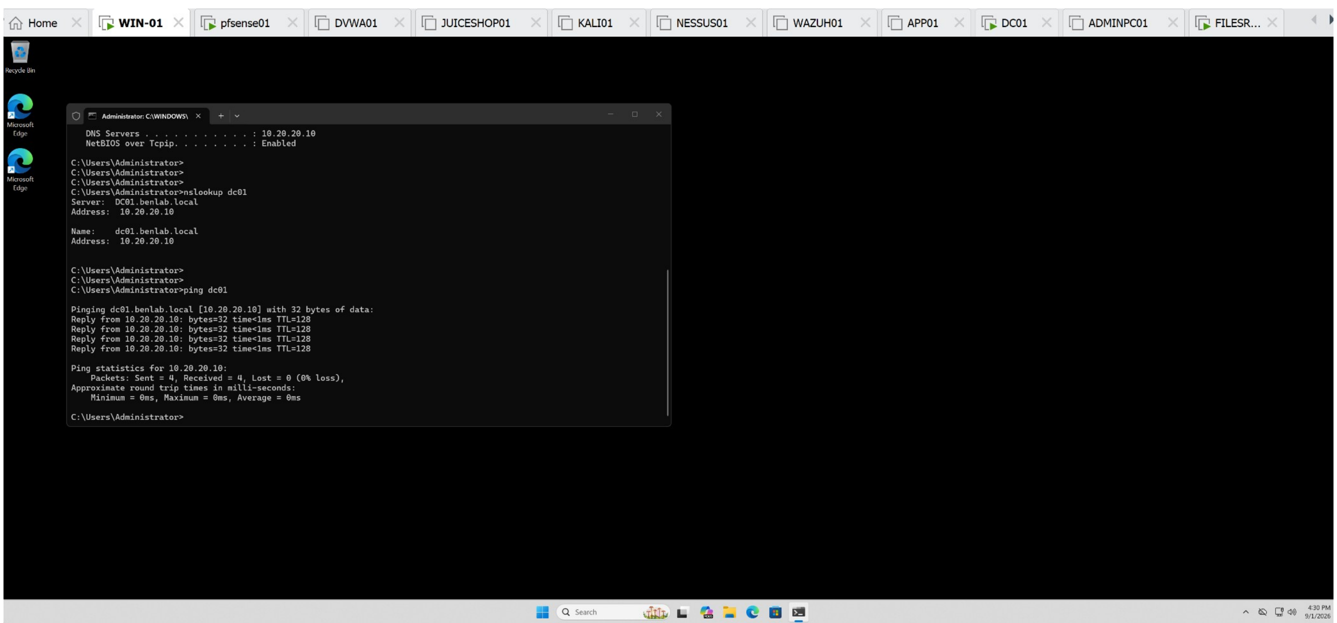
`10.20.20.10`



Internal connectivity was validated with:

`ping dc01`

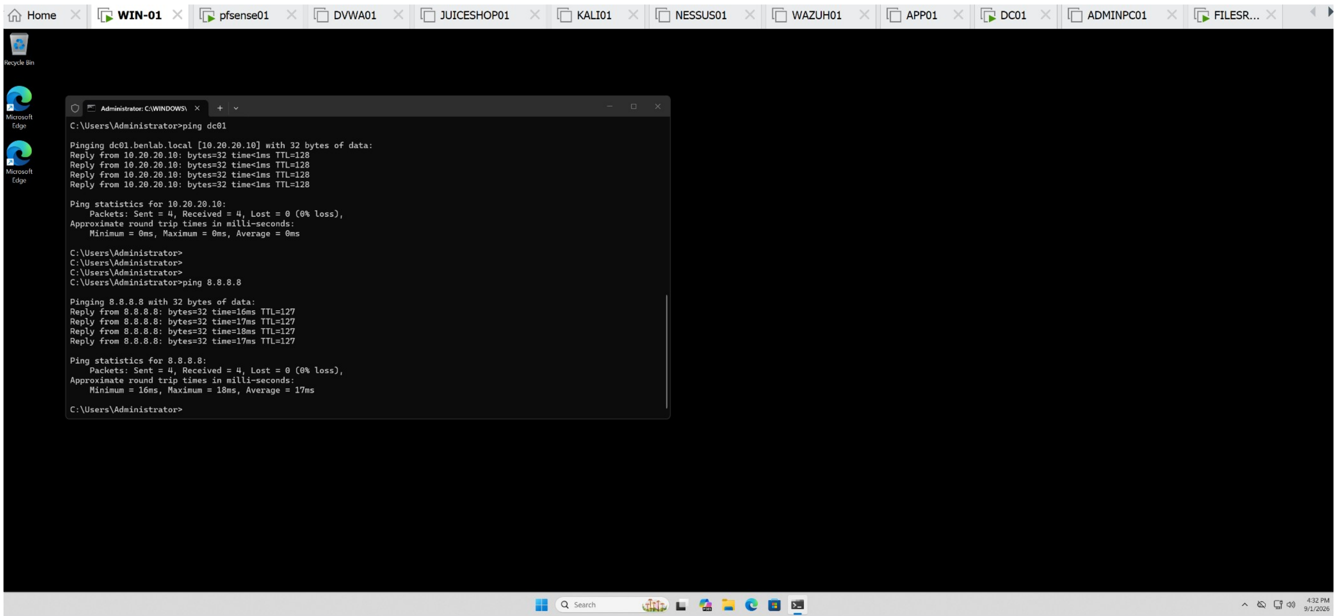
The test returned **4/4 successful replies with 0% packet loss**.



External IP connectivity was tested with:

`ping 8.8.8.8`

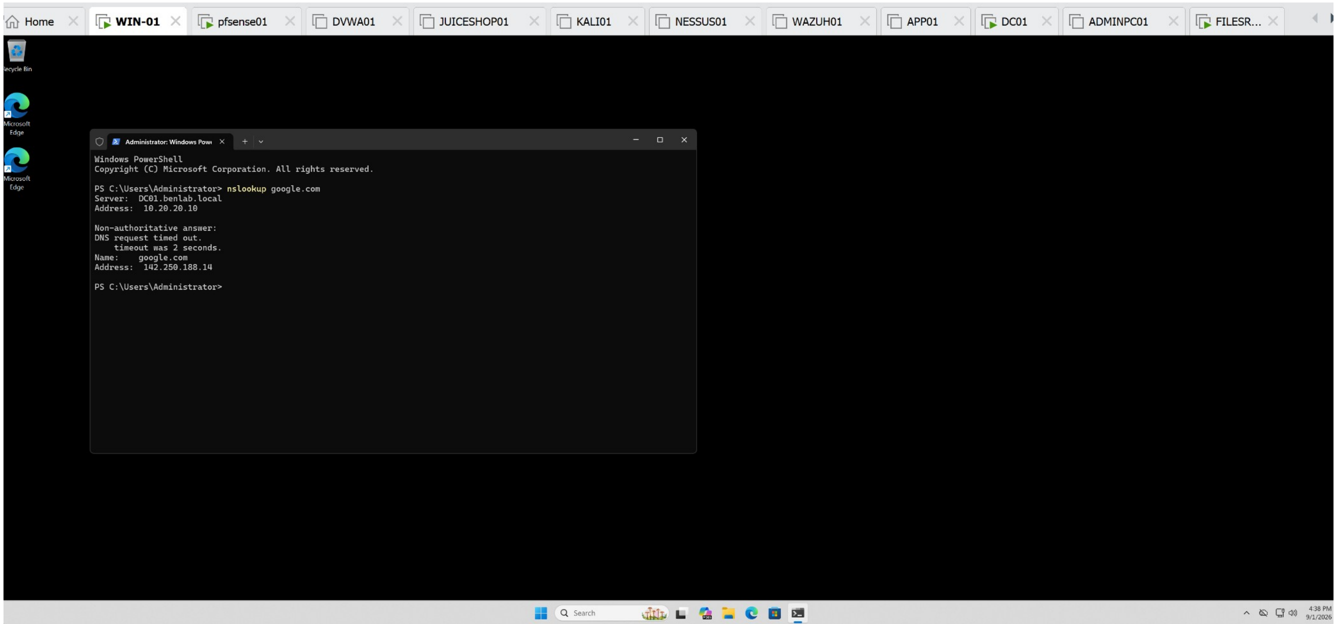
The test returned **4/4 successful replies with 0% packet loss**, confirming external routing through pfSense independently of DNS.



External DNS resolution was tested with:

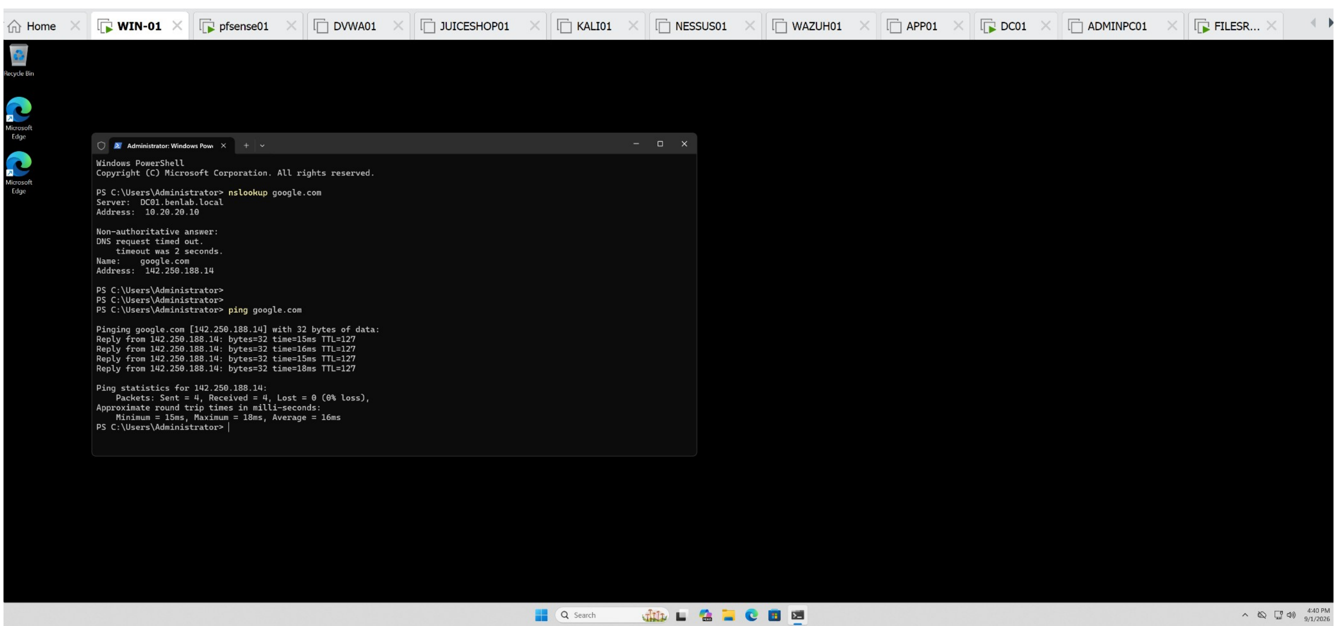
`nslookup google.com`

WIN01 queried DC01 at `10.20.20.10`, and `google.com` successfully resolved to a public IP address. A temporary two-second DNS timeout was observed before the successful response.



Finally:

`ping google.com`



successfully resolved `google.com` to `142.250.188.14` and returned **4/4 replies with 0% packet loss**.

Result:
Task completed successfully. End-to-end testing confirmed that WIN01 receives valid DHCP configuration from pfSense, uses DC01 for DNS resolution, communicates with internal domain resources, routes external traffic through pfSense, and successfully resolves and reaches Internet hosts.
